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From Latin to Romanian: Phonological rules

Pardess, David Forbes, Ph.D.

University of California, Los Angeles, 1990

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From Latin to Romanian:

Phonological Rules

A dissertation submitted in partial satisfaction of the
requirements for the degree Doctor of Philosophy
in Romance Linguistics and Literature

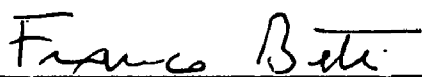
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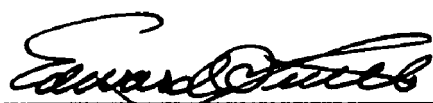
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1990

To the memory of
Elise Richter
and
to the peoples of Romania
in their ongoing quest
for freedom and prosperity.

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ABBREVIATIONS AND CONVENTIONS

A. Languages and Dialects

AR	Aromanian
CL	Classical Latin
DR	Daco-Romanian
Fr	French
IR	Istro-Romanian
It	Italian
LL	Late Latin
MR	Megleno-Romanian
Pt	Portuguese
Sp	Spanish

B. Abbreviations found in phonological rules

[acc]	primary stress
[ant]	anteriority
[back]	backness
[cons]	consonanticity
[cont]	continuity
[cor]	coronality
[del rel]	delayed release
[high]	height
[lat]	laterality
[low]	lowness
[long]	length
[nas]	nasality

[rep]	repressiveness
[round]	rounding
[son]	sonorance
[str]	stridency
[syl]	syllabicity
[tense]	tenseness
[voc]	vocalic
[voice]	voice

C. Segments

VOWELS = [-cons, +voc]

Inventory:

[a], [e], [ɛ], [i], [o], [u], [ə], [ɪ]

Tongue height:

[+high] = [i], [u], [ɪ]

[-high, -low] = [e], [ə], [o], [ɛ]

[+low] = [a]

Position:

[+back] = [a], [o], [u], [ə], [ɪ]

[-back] = [e], [ɛ], [i],

Length:

[+long] = [a:], [e:], [ɛ:], [i:], [o:], [u:]

[-long] = [a], [e], [ɛ], [i], [o], [u], [ə], [ɪ]

GLIDES = [-cons, -voc]

Inventory:

[e̞], [o̞], [y], [w], [h]

[+high] = [y], [w]

[-high] = [e̞], [o̞]

[+low] = [h]

[+back] = [o̞], [w]

[-back] = [e̞], [y], [h]

[w̥] and [y̥] = devoiced glides

CONSONANTS = [+cons, -voc]

Inventory:

[b], [β], [ɸ], [ɸ̥], [d], [dz], [f], [g], [ɣ], [j̥], [k],

[l], [ll], [λ], [m], [n], [nn], [ñ], [ŋ], [p], [r]*,

[rr]*, [s], [š], [t], [ts], [v], [x], [z], [ž], [ʒ]

(*[r] = flap; [rr] alveolar trill)

Anteriority and Corinality:

[+ant, -cor] = labials [p], [b], [m], [f], [v], [β]

[+ant, +cor] = alveolars [t], [d], [n], [nn], [s], [z],
[l], [ll], [r], [rr]

[-ant, +cor] = palato-alveolars [š], [ž], [č], [j̥]

= alveolo-palatals [ɸ] [ʒ]

[-ant, -cor] = velars [k], [g], [ɣ], [ŋ]

Other features used in categorizing consonants:

[+back] = velars

[+cont] excludes stops, nasals and affricates

[+del rel] excludes stops; used to distinguish affricates ([+del rel]) from stops ([-del rel])

[+high] = palato-alveolars, alveolo-palatals, velars

[-high] = alveolars, dentals, labials

[+rep] = distinguishes [rr] ([+rep]) from [r] ([-rep])

[+son] = vowels, nasals, glides, liquids

[-son] = stops, fricatives and affricates = obstruents

[+str] = fricatives and affricates

[-str] = excludes fricatives and affricates

D. Unilateral abbreviations of classes

C = all consonants; excludes V and W

K = obstruents; excludes V, W, R and S

L = liquids; includes [l], [ll], [ʎ], [r], [rr]

N = nasal consonants; includes [m], [n], [nn], [ɲ], [ŋ]

R = resonant consonants; includes N and L

S = sonorants; includes V, N, W, L

V = vowels; excludes C and W

W = glides

E. Conventions

Co any number of consonants

word boundary

\$ syllable boundary

> becomes (historically)

< is derived from (historically)

>> ultimately becomes (indicates that steps are missing)

<< is ultimately derived from
 / in the context
 // in the presence of
 * non-existent or unacceptable form
 [] enclose phonetic representations or distinctive features
 + presence of a feature
 - absence of a feature
 α the same value is to be chosen for each occurrence of
 () optional
 {} indicate an option (take one of the elements enclosed)
 — position of a segment
 ∅ null or zero
 X indicates an entity or context to be defined
 X_i X_i indicates that the two elements are identical
 X_i X_j indicates that the two elements are different
 ' stress

Classical Latin forms are written in capital letters when given in orthographic form.

Modern Romanian orthographic forms are underlined in the body of the text but not in the rules.

Phonetic representations are given in brackets in the body of the text but not in the rules or the examples following them.

English equivalents are enclosed within apostrophes.

E. Other abbreviations

f	feminine
g/d	genitive/dative
imperat	imperative mood
imperf	imperfect tense
m	masculine
n/a	nominative/accusative
pl	plural
pp	past participle
sg	singular

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ABSTRACT OF THE DISSERTATION

From Latin to Romanian
Phonological Rules

by

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Doctor of Philosophy in Romance Linguistics and Literature

University of California, Los Angeles, 1990

Professor Edward F. Tuttle, Chair

Latin and Romanian may be said to constitute the two extremes of a two thousand-year continuum. A basic assumption in this dissertation is that most changes in pronunciation are the reflections of rule changes that ultimately take place in the collective grammar of a speech community in the course of time. The goal was to state, as accurately as possible, what are thought to be the rules and to order them chronologically, or at least in a sequence in which they "feed or bleed" one another.

The method used varied according to the period studied. For the Balkan Romance period (first through sixth centuries A.D.) Carlos Otero's early rules for Hispano-Romance served

as a point of departure since all Romance dialects shared many rules as they diverged from their Latin parent. Each early rule had to be considered to verify that it would provide the proper output for Balkan Romance and a basis for rules in the Common Romanian period. Accordingly, various early rules had to be added, deleted, modified, or reordered in the sequence. The literature and grammatical commentary of this period ultimately provided the primary data for the formulation of these diachronic phonological rules. The work of Elise Richter in the early twentieth century provided the first chronology based on the literature.

Since there is no written record from the sixth to the sixteenth century, rules for the Common Romanian and Daco-Romanian periods had to be reconstructed internally. The reconstruction was based mainly on the work of two twentieth century Romanian linguists, Marius Sala and Emanuel Vasiliu, who collected and evaluated hypotheses for sound changes proposed by researchers in the nineteenth and twentieth centuries.

Results show that with few exceptions Romanian words inherited from Latin, as well as most of those borrowed from other stocks, underwent very regular developments. The low-level phonological rules formulated and sequenced in this dissertation, it is hoped, mirror historical changes in the phonological component of the grammar as Latin became

Romanian.

PREFACE

This dissertation is the result of my desire to better organize the data on the historical development of Romanian phonology found in the many handbooks written on the subject.

In the handbooks, from Densusianu's brilliant Histoire de la langue roumaine (1901) to the relatively complete and accurate Istoria limbii române of the Editura Didactică și Pedagogică (1978), sound changes are usually assigned to one of the traditional periods, i.e. Vulgar Latin, Balkan Romance, Common Romanian, Daco-Romanian, etc., but are not presented in the order in which they were thought to have occurred within each period. Consequently, it is difficult to keep track of the changes, and, when several hypotheses for the formulation or ordering of sound-change rules are presented, knowledge of the preceding changes forms a not insignificant part of the basis for choosing the best one. Another problem with the handbooks is that information contained in one frequently contradicts information found in another. I felt, therefore, that before it would be possible to do further meaningful and accurate research I

would have to formulate and order rules for Romanian much as Carlos Otero did in for Spanish and Galician-Portuguese in Evolución y revolución en romance, volumes 1 and 2.

This dissertation, then, is limited to the study of the historical phonology of modern standard Daco-Romanian, known in Romanian as 'limba literară'. Its purpose is to determine what regularities underlie the changes that occurred during the development of the pronunciation of Romanian and to formulate and order as accurately as possible phonological rules for the sound changes that are hypothesized to have taken place during its evolution.

Although the order of the rules presented in this analysis is intended to be a chronological order, it has turned out to be of necessity a hypothetical descriptive order, especially since after the Latin period there is no written corpus and little other linguistic data on Romanian until the early 16th century. However, as Artha Sue Loy discovered in her study of the development of French, there is "a close correspondence between the descriptive order and the chronological order of the rules." Otero also found that his descriptive order corresponded "to a certain degree" with the chronological orders of Loy, Elise Richter, Georges Straka and Ramón Menéndez Pidal.

It is to be hoped that the main product of this dissertation, an ordered list of phonological rules, will

facilitate further research on the history of the Romanian language, and perhaps even provide a basis for the investigation of the sequence of grammars of which they may be an expression.

I wish to express my gratitude and appreciation to my adviser Professor Edward Tuttle for his many suggestions, references and challenges, because of which the content of this dissertation has been greatly enriched, as well as for his unfailing support and encouragement, to Professor Carlos Otero for his instruction in the historical phonology of Spanish and Portuguese and generative theory as well as for several very valuable references, to Professor Carlos Quicoli for his references and for help in organizing the dissertation, to Professor Donca Steriade for verifying the accuracy of the Romanian examples, and to Professor Bruce Hayes for his references and advice, especially in regard to rule formulation.

My sincerest thanks go to the members of my immediate family for their encouragement and, most of all, their patience.

CHAPTER 1

INTRODUCTION

1.1 Sound Change: The Regularity Hypothesis

In the examples given below, the reflex of the original Latin *-kt-* is always the same in each of these Romance languages:

Late Latin	okto	faktu	nokte	lakte
Italian	otto	fatto	notte	latte
Portuguese	oito	feito	noite	leite
Spanish	ocho	hecho	noche	leche
French	huit	fait	nuit	lait
Romanian	opt	fapt	noapte	lapte

The discovery of regular correspondences such as these has inspired linguists interested in the historical development of languages to attempt to make generalizations about sound changes. One such generalization might read thus: "Latin *-kt-* became Romanian *-pt-*."

In the late 19th century, generalizations of this sort were given the name 'Lautgesetze' (sound laws) by a group of German linguists known as the 'Junggrammatiker' (neogrammarians) whose inspiration seems to have been the discovery that there was a regular correspondence between the obstruent systems of Germanic and other Indo-European languages. These correspondences, known as Grimm's Law and Verner's Law, led the neogrammarians to theorize that sound

laws have no exceptions. On the basis of this theory they formalized the regularity hypothesis: Change in pronunciation which is not conditioned by non-phonetic factors is regular and operates without exceptions at a particular time and in a particular speech community, with possible environmental restrictions. Certain changes, including dissimilation and metathesis, are exempt from this hypothesis. Apparent contradictions might be due to 1) an incorrect formulation of the sound change, 2) a later sound change somehow undoing the effects of the earlier change, 3) borrowing from other languages or dialects which, obviously, did not undergo the same sound changes in their histories, 4) borrowings from 'fast speech' into normal speech, such as *madam* > *ma'am*, 5) onomatopoeic words retaining their original imitative pronunciations, 6) the deliberate distortion of tabooed words, and, most importantly, 7) paradigmatic or other analogy, a morphological, non-phonetic factor which is by definition irregular (Hock 1986:34-51).

Although the regularity hypothesis was at first regarded as quite controversial, by the end of the 19th century it was generally accepted.¹

1.2 The Early Handbooks

The neogrammarian position expressed in the regularity

hypothesis was dominant during the early part of the 20th century when many of the handbooks on the histories of the pronunciations of various languages were published. Using the principle of regular sound change, the authors of these early handbooks were able to establish various stages in the development of a language, to reconstruct word forms for these stages, and to determine what types of changes occurred to produce these hypothetical forms.

Information used in this reconstruction was obtained from orthographic forms and patterns of rhyme and assonance found in texts, statements of grammarians, and the development of sounds and words borrowed into and from the language under consideration.

Sound changes were usually categorized as assimilation, dissimilation, subtraction, conditioned changes occurring only in specified environments, metathesis, addition, and isolative changes, which are not determined by environment.

The order of presentation of the sound changes varied from handbook to handbook. Some divided the book into chronological sections, discussed the development of each sound (usually all the vowels first, then all the consonants) in all possible environments in each era, and followed up with commentary and lists of exceptions.

Consequently, there was some sense of linear order to the changes. Although the ordering was meant to be purely

chronological, indicating that a particular change occurred during a particular time period because there was a change of spelling or of rhyme pattern attested in texts of that period, often it was actually a hypothetical descriptive order that was provided, especially when the texts did not clearly indicate when the change took place. A descriptive order may be characterized as one that is hypothesized on the basis of the data as logical and necessary but is not necessarily attested.

1.3 Elise Richter's Chronology

In her magnum opus Beiträge zur Geschichte der Romanismen: I, Chronologische Phonetik des Französischen bis zum Ende des 8. Jahrhunderts, 1934, Elise Richter went one giant leap beyond the handbooks by attempting to order chronologically the sound changes attested in post-classical Latin and in Gallo-Romance until the end of the 8th century A.D. using the method of internal reconstruction. In Richter's analysis, the relative chronology of the sound changes is established by applying a series of sound laws to the Latin base in such an order of application as to produce the forms attested or known to have existed at a particular time during the history of the language. In formulating her rules and order Richter realized that a relative chronology cannot be obtained from texts or the comments of grammarians

because sound changes usually take place long before they are documented; such information can only indicate the latest date by which a particular change occurred. Accordingly, some of her rules and ordering must have been hypotheses based on the data, and were, therefore, in reality descriptive rules.²

1.4 Structuralist Linguistics

The first major development in linguistics to challenge the neogrammarian position was the application of structuralist principles to the analysis of language.

1933 saw the publication of Leonard Bloomfield's Language, in which he set forth the major principles of American structuralism (also known as phonemic or taxonomic linguistics). Language dominated the field of linguistics for the next several decades. Before its publication, research on sound change had necessarily been conducted on the phonetic level. From the 1930's until the advent of generative theory, the effect of sound changes on the 'phonemic structure' of a language was generally the focus of what was considered meaningful and valuable research.

Some of the major aspects of the structuralist approach which distinguish it from both traditional phonetic and generative analyses are 1) a required locally determined phonemic transcription which meets the conditions of

linearity and invariance, 2) the following of certain procedures in the production of phonemic transcriptions, and 3) the disregard of syntactic information in an analysis.

1.5 Emanuel Vasiliu's Orientation and Contribution

To my knowledge, the only linguist to attempt to systematically chronologize the sound changes that took place in the development of Romanian per se is Emanuel Vasiliu.

Vasiliu's ultimate goal, however, in his Fonologia istorică a dialectelor dacoromâne, 1968, is to determine the effect of diachronic sound change on the traditionally conceived phonemic structure of each stage in the evolution of Latin to the Daco-Romanian dialects. To that end he hypothesizes that certain sound changes occurred during each of the generally accepted stages, i.e. Vulgar Latin, Balkan Romance, Common Romanian, and Daco-Romanian, seeks to establish the chronological order in which they occurred during the given stage, and then posits which sound changes caused which changes in the phonemic system of the stage.

So long as the researcher limits himself to saying that in Environment 1 the sound x becomes y, the ascertainment of the moment in which this change is produced is really nothing more than superfluous information. If, however, the researcher poses the problem of the phonemic status of sound y (resulting from sound x) he must necessarily establish whether at the moment of the change $x > y$ in Environment 1 there had occurred or not a series of other sound changes to decide if by means of $x > y$ there appears a new phoneme,

a new variant, whether an instance of neutralization is produced or a total confusion of the two phonemes (Vasiliu 1968:21) [translation mine].

To illustrate. In order to be able to establish the evolution of the Latin vowel system with five vowel phonemes to the Romanian system with six and later seven vowel phonemes, it would be necessary to establish the chronology of all changes which result in the new phones [ə] and [ɪ] and also take inventory of all the situations in which the Latin phones are preserved unchanged. Without this chronology it would be impossible to say which is the change that determined the appearance of the phoneme /ə/ or which is the change which caused the phoneme /ɪ/. If the relative chronology of these 'decisive' phonetic changes is not established (or cannot be established), as Vasiliu sees it, one merely succeeds in producing a formulation of trivial value, such as 'the totality of the sound changes undergone by the vowels of Latin determined the appearance of the Romanian vowel system'. For Vasiliu, the trivial character of this statement is due to the fact that investigation has not led to the establishment of the way in which the sound changes under consideration affected the phonemic system and structure (Vasiliu 1968:21 ff.).

In this way, Vasiliu attempts to do what generativists call 'investigating the effects of the rules on the sequence of grammars'. His analysis, however, is very difficult to

follow since the logically established 'feed or bleed' relationships for his sound change rules are scattered throughout the book. Moreover, Vasiliu has admittedly omitted several interesting sound changes from his analysis.

1.6 Generative Phonology

In the generative model of language, which has been dominant for the greater part of the last three decades, linguistic research is subject to a far more precise theory of grammar than it was under structuralism and is formalized in terms of modern mathematical notions. As a result, there is no separate phonemic level in the structuralist sense.

The purpose of generative phonology (and generative grammar as a whole) is to characterize native speakers' (subliminal) knowledge of the grammar of their language. This knowledge, it is assumed, permits speakers not only to assert that what is phonetically identical is grammatically different (and vice versa), but also to make judgments concerning the correctness of given utterances, and perhaps most importantly, to acquire their own native language.

In the view of generative grammar, this grammatical knowledge is characterized by a system of rules which relate different levels (syntactic, semantic, phonologic, etc.) to each other and which ultimately map meaning into phonetic structure. Associated with this rule system is a lexicon which contains the lexical items of the language, along with the various features or markings which determine or govern their interaction with the rule system (Hock 1986:248).

The generativist school uses the notational system of binary distinctive features, proposed by the structuralist Roman Jakobson and reworked and updated many times in the last 20 or so years, in order to provide a more general

formulation of the phonetic facts than is possible with the use of traditional phonetic or phonemic transcriptions. It also allows rules to be collapsed, which results in a more economical description.

Another contribution of generativism is the explicit inclusion of a simplicity criterion within its framework, which provides a means of selecting a particular rule over others which will correctly account for a given body of data. According to the simplicity criterion, the description that is the most economical is the best. The use of binary distinctive features facilitates measuring simplicity.

The phonological similarities and differences between two diachronically distinct states of the same language can be described in terms of the similarities and differences in two synchronic descriptive rule systems such as two contemporary dialects of the same language. One system may contain a synchronic rule that the other lacks (or may restrict its application more or less narrowly); one system may differ from the other in that the same set of rules will apply in a different order in the one system from the order in which they apply in the other.

According to Halle (1962) the basic type of sound change, namely innovatory change, is reflected in the generative model by the addition of a new phonological rule at the end of the phonological component. The addition of the new rule has the effect of systematically modifying the phonetic representations

generated by the grammar so that they differ from those generated by the previous grammar in which the rule is lacking. The difference in output must obviously be small enough not to disrupt communication between possessors of the old and the new grammars. This type of change may typically be attributed to the adult speaker, who is capable of modifying his speech in limited ways. The output of this modified grammar will then serve as data for the language learner of the following generation who, it is claimed, has as a child the innate capability of constructing from the data to which he is exposed the 'optimal' or simplest possible grammar. This means that the added rule will be retained in the grammar of the next generation only if a simpler grammar cannot be constructed which will generate the same output. If, however, this is possible the grammar will be restructured so as to account for the same output more simply. Broadly speaking, the new rule will remain as a functional rule in the synchronic grammar of the following generation only if it results in a morphological alternation, otherwise its effects will be absorbed into the lexicon causing restructuring of the relevant underlying representations (Bynon 1977:114).

Generative theory at first attempted to broaden the range of sound change by claiming that it could also be morphophonemically conditioned, that is to say that rule addition could operate directly on the underlying lexical representations (Halle 1962:68ff.) (Bynon 1977:118).

Subsequently this position was abandoned. It is now generally agreed that new rules may be added at one point only, namely at the end of the phonological rules (Kiparsky 1972; King 1973). Accordingly, the concept of 'rule insertion' at a higher level of the grammar has been rejected.

1.7 Studies Based on Early Generative Theory

In her doctoral dissertation entitled Historical Rules in the Development of Modern French from Latin, 1966, Artha

Sue Loy's goal is "to present a set of ordered phonological rules which represent the major sound changes that occurred during the development of the French language which form a part of the phonological component of a transformational [generative] grammar."

Using data from traditional handbooks and the method of internal reconstruction and adhering strictly to early generative theory, she obtains a set of rules which is very similar to that of Richter. Loy states that her order, too, is quite like Richter's, but that it is difficult to compare the two analyses mainly "because not all of the rules in a transformational phonology reflect the order in which they might have appeared chronologically" and cites Postal's well known rule of epenthesis in Mohawk as an example. This particular difficulty of comparison probably no longer exists since the concept of rule insertion at a higher level in the grammar has been abandoned.

Apart from the theory and notation that separate them, a major difference between Loy's generative analysis and a traditional analysis such as Richter's turns out to be that in a generative analysis rules which applied at different points chronologically but which share the same type of change are collapsed or combined in order to express an underlying generalization common to the rules in question. This may result in their defying an attested chronological

order.

In his book Evolución y revolución en romance, volume 1, 1971, Carlos Otero includes a list of generative 'procesos fonológicos' which trace the development of Spanish from Latin. In volume 2, 1976, he gives further rules for the development of Galician-Portuguese, Tuscan, French, and a few hints about rules and ordering for Romanian.

In Otero's analysis, all rules except those identified as permanent ('fuera de serie') or recurrent seem to be added at the end of the phonological component, thereby making it reasonable to equate his descriptive order with a hypothetical chronological order.

Otero further emphasizes that the product of his analysis is limited to a list of processes or rules that pave the way for research on their effect on the series of synchronic grammars. "Téngase muy en cuenta que no ha sido propósito de este estudio investigar la secuencia de las gramáticas, aunque sí desbrozar el camino para esa investigación (mucho más exigente) (Otero 1971:292)."

1.8 Generative Phonology and Historical Phonology Today

According to Hock, early generativists made several assertions or predictions about language change which may be in need of revision. For example, it is not certain that

all linguistic change leads to simplification of a grammar; on the contrary, there seem to be many changes which complicate a grammar. Another tenet is that all linguistic change is grammar change; it seems, however, that even changes which may have far-reaching grammatical implications may be motivated by surface patterns instead of the grammatical rule system. Also, it is difficult to account for leveling and the less systematic analogical processes as grammar change (Hock 1986:278).

Although radical changes such as these have led some linguists to question the validity and the usefulness of the generative approach for historical phonology, Hock feels that it is neither desirable nor necessary to consider discarding it. The updated generative model described in 1.6 seems to be adequate for the purposes of this dissertation.

1.9 The Mechanism of Linguistic Change

Perhaps the most widely accepted theory of linguistic change today is that propounded by William Labov, who sees social motivation as the most powerful cause of linguistic change. His theory has been summarized by Hock as follows:

- (i) The basis for linguistic change lies in the same ever-present low-level variability of ordinary speech that was noticed by the nineteenth-century phoneticians and by the neogrammarians. (ii) For reasons which linguistically appear to be completely arbitrary, a given phonetic variable, out of the many which are found

in ordinary speech, is selected for being socially significant as a marker of group identification. (iii) As a consequence of this sociolinguistic marking, the variable ceases to be a mere 'performance' variant and attains grammatical significance. It becomes a matter of what generativists call 'competence'. This is shown for instance by the fact that over- or under-use of the variable is noticed by speakers much as a speech error or non-native 'accent' is registered. (iv) Having thus attained grammatical significance, the variable must be accounted for by the grammar of the (social) dialect, in terms of a variable rule which is conditioned both linguistically (in terms of phonology) and socially. (v) This rule in turn, because it is a rule (i.e. 'generalization') and also because of the social reasons that gave rise to it, tends to be generalized or extended to (a) new environments, (b) new word classes, and (c) occasionally to structurally or functionally similar segments. The extension to new word classes permits the change to be completely generalized. And the extension to structurally or functionally similar segments allows for the possibility of chain shifts. (vi) Social groups hardly ever are completely exclusive of each other, but commonly have overlapping membership; cf. the overlapping group memberships defined by a term like 'White Anglo-Saxon Protestant Professor of Forensic Medicine'. As a consequence, the social parameters for the generalization of the variable rule may be expanded to ever larger groups. Along the way, the social marking of the variable may well undergo reinterpretation. (vii) To the extent that there are no opposing social pressures which might block complete generalization of the variable rule, it may eventually spread through the whole lexicon and throughout the whole speech community. (viii) Because generalization usually is not in terms of individual words but of variable rules, the ultimate outcome is said to most commonly consist in (virtually) complete regularity. (ix) At this point of complete generalization, then, the variable becomes part of the normal, socially unmarked inventory of the speech community and, as a consequence, the variable rule accounting for it ceases to operate. That is, in generative terminology, there is restructuring. (x) The fact that this pattern of sound change implementation is observable in all studies of change in progress suggests that also other, completed changes were implemented in this fashion (Hock 1986:648-9).

1.10 Theory and Method in this Dissertation

Like most historical linguists I feel it is reasonable to assume that each sound change begins as an alternate pronunciation which eventually predominates and, with the passing of several generations, may entirely replace the older form. The most convincing explanation of the mechanism of this sort of change is that of Labov as outlined above.

Diachronic Phonological Rules

With the generativists I assume that a grammar is basically a set of rules having at least a syntactic, a semantic and a phonological component, which children naturally acquire as they acquire the lexical items of their language in context. At any given time the rules for the phonological component are established and ordered. Each speaker knows these rules tacitly and applies them as the need arises. The phonetic realization of noun pluralization in English is a good example.

Rule addition or other rule change probably takes place at the end of the list of established and ordered rules of the phonological component, which is very near the surface level of a language. If it causes a split (i.e. a morphological alternation), it remains at the phonological level as a rule. If it simply replaces an older sound, succeeding generations acquire the changed form as a part of a lexical item at a much deeper level within the language

model, i.e. restructuring at the lexical level occurs.

In order to chart the course of the history of the pronunciation of a language it is necessary to find patterns of sound change, make generalizations and formulate rules to express these generalizations. The environment in which each change takes place must also be specified.

Obviously, phonological rules are an abstract representation of a concrete reality, i.e. pronunciation change. To quote Vasiliu [translation mine]:

The only immediate and real given that the researcher operates with is the fact that: a) sound x of Language 1 corresponds to sound y of Language 2 (within a specified context or independently of any context) and b) that this correspondence is repeated a sufficient number of times (not necessarily always); this repeatability of the correspondence makes it possible to characterize it as regular.

The concept "sound change" is not an immediate and observable reality; such a concept does nothing more than explain a detected correspondence in a strictly empirical way. A formula such as $x > y$ is nothing more than a mental construct designed to explain a fact, i.e. that a correspondence is regular. In this sense then, a phonological rule represents the explicative hypothesis of a reality, and not the reality itself. Therefore, the formulization of series of phonetic changes does not represent the recording of facts, but rather the formulation of propositions which refer to facts. The consequence of this state of affairs is that a sound change (or a series thereof) is subject to testing, as is any hypothesis, a) on the basis of its explicative capacity, i.e. is it more general than another competing hypothesis? does it explain a greater number of facts than a competing hypothesis? b) on the basis of its ability to relate to other hypotheses i.e. does it accord with another hypothesis which refers to another domain of facts or to a wider domain better than a competing hypothesis does? c) on the basis of simplicity, i.e. does it explain the same phenomenon in a simpler fashion than does a competing hypothesis (Vasiliu 1968:15 ff.)?

A rule must be expressed in the generative format if it is ultimately to interface with the other components of the generative model. Aspects of the distinctive feature notational system are still being hotly debated by linguists. The one used here is fairly basic and conservative but is relatively up to date and adequate for the purposes of this dissertation. Unfortunately, distinctive feature notation is difficult for all but the initiated to decipher, so for the sake of general accessibility I feel it is advisable to state a rule in three formats: with phonetic characters, with the binary feature system if there is an interesting or useful generality to capture, and with an additional prose explanation if necessary for maximum comprehensibility.

In general, all other things being equal, application of the simplicity criterion to rule formation and ordering seems to be the most reasonable way of proceeding unless such application would violate a known or reasonably well hypothesized chronological sequence. Also, adherence to the principles of naturalness and uniformitarianism is necessary in order to limit the explicative power of rules to take into account what it is natural to expect in the way of human language change. For a concise discussion of naturalness see Wolfram and Johnson (1982:158 ff). The application of uniformitarian principles to linguistics is

discussed in Christy (1983:passim).

In historical linguistics, every aspect of work draws on our knowledge of what is universal in language--what is possible and likely in a natural language, what processes of change may be assumed as plausible.

At every step toward the recovery of earlier structure we must ask ourselves whether the reconstructed stage is possible in terms of what we know to be generally true of natural languages--i.e to be linguistic universals or near-universals (King 1969:15).

Rule Ordering

It is generally accepted that rule ordering is extremely important in historical phonology. Actually, rule formulation and rule ordering must be thought out in close conjunction with each other.

Some rules turn out to have no effect on other rules; their place in the overall order seems relatively unimportant provided that they occur in the proper era and accord with any philological evidence there might be. On the other hand, some rules 'feed' or 'bleed' other rules. Their place in the order, relative to the rules they feed or bleed, is obviously essential.

Halle (1962) made a case for a dichotomy between chronological ordering and descriptive ordering as described in 1.2. I think almost all ordering in which the rules feed or bleed one another, even where it exists in the manuals written in the early twentieth century and is intended to be chronological, may actually be descriptive.

In historical phonology our ultimate goal should be to

establish the simplest rules that will account for the data and order them as they could have reasonably occurred chronologically. To my mind, this is the only worthwhile kind of descriptive ordering.

Data

Obviously, data is necessary for hypothesizing sound changes. Consider the following: If we have only the language itself and no information about a parent language or genetically related (sister) languages or dialects, we cannot proceed with a diachronic analysis. If we have the language and related languages and dialects but not the parent, we can proceed backwards and by using the comparative method arrive at a hypothetical parent form. That is what 19th-century linguists did to reconstruct Proto-Indo-European. When the parent is known as is the language and its dialects and sisters and there is a more or less constant flow of literature and grammatical commentary during the entire continuum, philological information provides the principal data for the analysis. Such is the case with standard Spanish, French, Italian and Portuguese. When the language and its dialects and sisters are known, as well as the parent but there is no literature or other philological data for a long interval, the undocumented period must be reconstructed. Such is the case with Romanian.

Method

Although it is assumed that the Romance languages developed from the spoken language of the people, my practice has been to use Classical Latin, which was fixed in form by the beginning of our era, as a point of departure since classical forms of languages tend to be more conservative phonologically. Although some points are still debated, there seems to be enough agreement to justify the version of the CL sound system outlined in Chapter 2. However, when the Romanian reflex indicates that derivation from CL is unlikely, I do not hesitate to look for a source in post-Classical Latin or elsewhere.

Post-Classical Latin has been studied extensively and reconstructed from various fragmentary sources such as the literary lapses of Roman authors, the deliberate use of non-standard forms by writers seeking comic effect, the observations and censures of Roman grammarians, graffiti and epigraphical remains, the works of Christian writers, Latin words borrowed by the peoples with whom the Romans came in contact, and the evidence furnished by Latin's Romance descendents.

Although most of the early sound changes in post-Classical Latin are shared by the majority of the Romance languages, its evolution in the eastern provinces up till the sixth century has been designated Balkan Romance by

Vasiliu and others.

Otero's early rules, most of which are more or less pan-Romance in scope, have provided the basis for rules for Balkan Romance in this dissertation. The Italian translation of Väänänen's Introduction au latin vulgaire has been the major source of commentary and attestations.

It has been more difficult to reconstruct Common Romanian and early Daco-Romanian since there are no texts written in Romanian until A.D. 1521. It has, however, been possible to hypothesize changes and order them within these periods, by comparing the four major Romanian dialects and also by using the method of internal reconstruction.

The four major Romanian dialect groups include Daco-Romanian, presently spoken in Romania and the Moldavian SSR by over thirty million people, and Aromanian, Megleno-Romanian and Istro-Romanian, spoken south of the Danube by very small populations whose ancestors were separated from the main body of speakers at various times after the Common Romanian period.

Many hypotheses have been advanced concerning the sound changes which occurred during the Common Romanian and Daco-Romanian periods, the best supported of which have been collected by Vasiliu and Marius Sala, whose works I have frequently consulted in the preparation of this dissertation. For other phenomena not included in Otero, Vasiliu or Sala I

have consulted handbooks, especially Istoria limbii române, 1978, and use the principles above plus logical inference to hypothesize the changes.

Once the rules are tentatively formulated and ordered using data, commentaries, and the suggestions provided by previous research, they are subject to verification by actually deriving Romanian reflexes from their Latin etyma (see Chapter 8).

Finally, since all rules in this dissertation are hypotheses, they stand to be criticized, amended, or even replaced in response to further research.

The fact that we reconstruct protosystems must not lead us to assume that our reconstructions are either definitive or complete. Reconstructions are not once and for all discoveries which will retain their validity indefinitely but, like any analysis, are always open to revision as the result of theoretical advances of the acquisition of new data (Bynon 1979:73).

CHAPTER 2

SOME PHONOLOGICAL INFORMATION ABOUT CLASSICAL LATIN

2.1 Introduction

The phonology of Latin, like that of any other natural language, was in a constant state of flux. However, since we must begin at some definite point of departure, the most obvious recourse seems to be to choose what is perceived to be the sound system of Classical Latin, some important points of which are outlined in this chapter. Ever since CL was fixed at the beginning of the Christian era, grammarians like Probus were constantly using it as a standard with which to compare what they saw as the decadent contemporary idiom. Since we have only the written word and CL grammarians' commentary, this analysis begins of necessity on an abstract level.

2.2 Vowels

As is well known, CL had five vowel phonemes:

	front	central	back
high	i		u
mid		e	o
low		a	

each of which could be qualitatively short or long.

	i	i:	e	e:	a	a:	o	o:	u	u:
syllabic	+	+	+	+	+	+	+	+	+	+
sonorant	+	+	+	+	+	+	+	+	+	+
consonantal	-	-	-	-	-	-	-	-	-	-
high	+	+	-	-	-	-	-	-	+	+
low	-	-	-	-	+	+	-	-	-	-
back	-	-	-	-	+	+	+	+	+	+
tense	+	+	+	+	+	+	+	+	+	+
round	-	-	-	-	-	-	+	+	+	+
long	-	+	-	+	-	+	-	+	-	+

According to some scholars the long variant of each vowel was tense, and the short variant lax. The Roman grammarian Varro, however, seems to indicate that in the Classical period the lax pronunciation of short [i] and [u] was considered rustic. For the purposes of this study, we shall consider that all CL vowels are [+tense] and that changes in vowel quality occurred later.

Vowel length was phonemic in CL:

<u>ANUS</u>	[á-nus]	'old woman'
<u>ĀNUS</u>	[á:-nus]	'ring'

There were several cases in which unstressed short [i] and [u] were non-syllabic, i.e. [y] and [w]:

1) when they functioned as the non-syllabic element in the three CL diphthongs AE, OE and AU:

<u>CAELUM</u>	[káy-lum]	'sky'
<u>FOETUS</u>	[fóy-tus]	'offspring'
<u>TAURUS</u>	[táw-rus]	'bull'

2) when they were syllable initial and preceded another vowel:

JURĀRE [yu-rá:-re] 'to judge'

VENĪRE [we-ní:-re] 'to come'

3) when U functioned as the co-articulaed labio-velar segment [w] in the combinations QU and GU:

AQUA [á-kwa] 'water'

LINGUA [líŋ-gwa] 'tongue'

All other contiguous vowels were syllabic:

PUTEUS [pú-te-us] 'well'

FOLIA [fó-li-a] 'leaf'

Non-syllabic I was geminated in intervocalic position:

MAIOR [má-y-or] 'greater'

EIUS [ey-yus] 'of him, her, it'

2.3 Consonants

CL had 15 consonant phonemes:

	p	b	m	f	t	d	n	l	r	s	k	g	h
syllabic	-	-	-	-	-	-	-	-	-	-	-	-	-
sonorant	-	-	+	-	-	-	+	+	+	-	-	-	-
consonantal	+	+	+	+	+	+	+	+	+	+	+	+	+
anterior	+	+	+	+	+	+	+	+	+	+	-	-	-
coronal	-	-	-	-	+	+	+	+	+	+	-	-	-
high	-	-	-	-	-	-	-	-	-	-	+	+	-
low	-	-	-	-	-	-	-	-	-	-	-	-	+
back	-	-	-	-	-	-	-	-	-	-	+	+	-
continuant	-	-	-	+	-	-	-	+	-	+	-	-	+
strident	-	-	-	+	-	-	-	-	-	+	-	-	-
del release	-	-	+	+	-	-	+	+	-	+	-	-	+
voiced	-	+	+	-	-	+	+	+	+	-	-	+	-
nasal	-	-	+	-	-	-	+	-	-	-	-	-	-
lateral	-	-	-	-	-	-	-	+	-	-	-	-	-

The sound [ŋ] occurred in two cases:

1) as a variant of [n] before a velar consonant

LINGUA	[líŋ-gwa]	'tongue'
--------	-----------	----------

PUNCTUM	[púŋk-tum]	'point'
---------	------------	---------

2) as a variant of [g] before [n]

LIGNUM	[líŋ-num]	'wood'
--------	-----------	--------

SIGNUM	[síŋ-nus]	'sign'
--------	-----------	--------

Some consonants could occur geminated. Consonant length could be phonemic:

ANUS	[ánu:s]	'old woman'
------	---------	-------------

ANNUS	[ánnu:s]	'year'
-------	----------	--------

The letter X usually represented the sequence [ks], but before a voiced consonant it was pronounced [gz].

2.4 Stress

Ancient sources indicate that in CL the accent consisted of a change in tone, presumably a raising of the voice on a given syllable. It is now generally agreed, however, that these sources were influenced by a statement about Greek, and that in Latin the accent was primarily expiratory, i.e. one of stress or intensity (Hammond 1976:72).

CHAPTER 3

SOME PHONOLOGICAL INFORMATION ABOUT MODERN STANDARD ROMANIAN

3.1 Vowels

Romanian has seven vowel phonemes:

	front	central	back
high	i	ɨ	u
mid	e	ə	o
low		a	

There are no long vowels in Romanian. The following is a chart of the distinctive features of these vowels:

	i	e	ə	ɨ	a	o	u
syllabic	+	+	+	+	+	+	+
sonorant	+	+	+	+	+	+	+
consonantal	-	-	-	-	-	-	-
high	+	-	+	-	-	-	+
low	-	-	-	-	+	-	-
back	-	-	+	+	+	+	+
tense	+	+	+	+	+	+	+
round	-	-	-	-	-	+	+
long	-	-	-	-	-	-	-

Note that [ə] is [+tense] in Romanian.

When contiguous to another vowel, i, u, e, and o are usually non-syllabic. So it could be said that Romanian has four glides: [y], [w], [ɛ], and [ɔ].

Examples:

piatră [pyá-trə] 'stone'

<u>haină</u>	[háy-nə]	'jacket'
<u>sau</u>	[sáw]	'or'
<u>guaşă</u>	[gwášə]	'gouache'
<u>beat</u>	[beát]	'drunken'
<u>foame</u>	[fɔá-me]	'hunger'

Contiguous vowels occurring in foreign words and neologisms are sometimes heterosyllabic, i.e. in hiatus:

Examples:

<u>liana</u>	[li-á-nə]	'liana'
<u>nuclear</u>	[nu-cle-ár]	'nuclear'

Noteworthy is the hiatus that occurs between a and u in words inherited from Latin in stressed, non-final position:

Examples:

<u>aur</u>	[á-ur]	'gold'
<u>taur</u>	[tá-ur]	'bull'
<u>aud</u>	[a-úd]	'I hear'

In Latin the combination AU in the same words always represented a diphthong:

Examples:

AURUM	[áw-rum]	'gold'
TAURUS	[táw-rus]	'bull'

Romanian is rich in diphthongs and triphthongs, the most important of which are:

[ɛa]	<u>deal</u>	[deál]	'hill'
[ɛo]	<u>vreo</u>	[vreó]	'some'

[ya]	<u>piatră</u>	[pyátrə]	'stone'
[ye]	<u>miel</u>	[myel]	'lamb'
[ɔa]	<u>foame</u>	[fɔáme]	'hunger'
[wa]	<u>guașă</u>	[gwášə]	'gouache'
[əw]	<u>rău</u>	[ráw]	'bad'
[ow]	<u>cadou</u>	[kadów]	'gift'
[iw]	<u>fiu</u>	[fíw]	'son'
[ɪw]	<u>riu</u>	[ríw]	'river'
[ay]	<u>haină</u>	[háynə]	'jacket'
[ey]	<u>vei</u>	[vey]	'you will sg'
[əy]	<u>măicuță</u>	[məykútsə]	'sister (nun)'
[ɛy]	<u>cîine</u>	[kíyne]	'dog'
[oy]	<u>doi</u>	[dóy]	'two'
[uy]	<u>cuib</u>	[kúyb]	'nest'
[ɔay]	<u>frantúzoaică</u>	[frantsuzoáykə]	'French woman'
[ɛay]	<u>spuneai</u>	[spunɛáy]	'you said imperf'
[ɛaw]	<u>spuneau</u>	[spunɛáw]	'they said imperf'
[yay]	<u>contribuiai</u>	[kontribuyáy]	'you contributed sg'
[yaw]	<u>iau</u>	[yáw]	'I take'
[yew]	<u>eu</u>	[yew]	'I'
[yey]	<u>iei</u>	[yey]	'you take sg'

3.2 Unstressed, Word-final i

Variants of unstressed i occur word-finally
as the plural marker for masculine nouns and adjectives

lup 'wolf'

lupi 'wolves'

metru 'meter'

metri 'meters'

cîine 'dog'

cîini 'dogs'

slab 'thin sg'

slabi 'thin pl'

mic 'small m sg'

mici 'small m pl'

frumos 'pretty m sg'

frumoși 'pretty m pl'

and certain feminine nouns and adjectives

biserică 'church'

biserici 'churches'

țară 'land'

țări 'lands'

floare 'flower'

flori 'flowers'

as the second person singular marker in most verb forms

eu cînt 'I sing'

tu cînți 'you sing'

eu cîntam 'I sang'

tu cîntai 'you sang sg'

at the end of the second person plural marker of most
verbs forms

voi cîntați 'you sing'

voi ați cîntat 'you sang'

in some indirect object clitics

îmi 'to me, for me'

îți 'to you, for you'

îi 'to him, for him, etc.'

își 'to himself, for himself, etc.'

and in some adverbs, conjunctions, and other invariable
words

atunci 'then'

ori 'or'

In all the above cases unstressed, word-final i has

varying phonetic values:

a) [i] after a consonant plus a liquid

<u>metri</u>	[metri]	'meters'
<u>codri</u>	[kódri]	'forests'
<u>umpli</u>	[úmpli]	'you fill sg'
<u>umbli</u>	[úmbli]	'you go sg'

b) [y] after a vowel

<u>nai</u>	[náy]	'pan pipes'
<u>vei</u>	[vey]	'you will sg'
<u>noi</u>	[nóy]	'we'
<u>pui</u>	[púy]	'chick'
<u>hăi</u>	[háy]	'hey'
<u>întîi</u>	[întîy]	'first'

c) [y] after voiced non-velar, non-palatal consonants

<u>slab</u>	[sláb]	'weak m sg'
<u>slabi</u>	[sláby]	'weak m pl'
<u>slav</u>	[sláv]	'Slav m sg'
<u>slavi</u>	[slávy]	'Slav m pl'
<u>final</u>	[finál]	'final m sg'
<u>finali</u>	[finály]	'final m pl'
<u>pom</u>	[póm]	'fruit tree'
<u>pomi</u>	[pómy]	'fruit trees'
<u>bun</u>	[bún]	'good m sg'
<u>buni</u>	[búny]	'good m pl'

<u>mare</u>	[máre]	'big m sg'
<u>mari</u>	[máry]	'big m pl'

d) [ɣ] after voiceless non-velar, non-palatal consonants

<u>ciorap</u>	[coráp]	'sock'
<u>ciorapi</u>	[corápy]	'socks'
<u>crap</u>	[kráp]	'carp'
<u>crapi</u>	[krápy]	'carp pl'
<u>pantof</u>	[pantóf]	'slipper'
<u>pantofi</u>	[pantófy]	'slippers'
<u>cartof</u>	[kartóf]	'potato'
<u>cartofi</u>	[kartófy]	'potatoes'
<u>drăguț</u>	[drəgúts]	'sweetheart'
<u>drăguți</u>	[drəgútsy]	'sweethearts'

e) combines with [t] to create [tsy]

<u>cît</u>	[kît]	'how much m'
<u>cîți</u>	[kíttsy]	'how many m pl'
<u>bărbat</u>	[bərbát]	'man'
<u>bărbați</u>	[bərbátsy]	'men'

f) combines with [d] to create [zy]

<u>văd</u>	[vəd]	'I see'
<u>vezi</u>	[vezy]	'you see sg'
<u>brad</u>	[brad]	'fir'
<u>brazi</u>	[brazy]	'firs'

g) combines with [s] and [z] to create [ɕ] and [ʒ]

<u>urs</u>	[urs]	'bear'
<u>urși</u>	[urɕ]	'bears'
<u>rus</u>	[rus]	'Russian m sg'
<u>ruși</u>	[ruɕ]	'Russian m pl'
<u>obraz</u>	[obráz]	'cheek'
<u>obraji</u>	[obráʒ]	'cheeks'
<u>mînz</u>	[mɨnz]	'colt'
<u>mînji</u>	[mɨnʒ]	'colts'

h) combines with [k] and [g] to create [ç] and [j]

<u>mic</u>	[mik]	'small m sg'
<u>mici</u>	[miç]	'small m pl'
<u>copac</u>	[kopák]	'tree'
<u>copaci</u>	[kopáç]	'trees'
<u>rog</u>	[rog]	'I beg'
<u>rogi</u>	[roj]	'you beg sg'
<u>drag</u>	[drag]	'dear m sg'
<u>dragi</u>	[draj]	'dear m pl'

3.3 Vowel alternations

The following vowels and diphthongs alternate synchronically in Romanian in words that inflect: [e/ɛa], [o/ɔa], [ə/e], [a/e], [a/ə], and [o/u]. There are two sorts of vowel alternation in Romanian: metaphonic and accentual. Metaphonic vowel changes occur in nouns, adjectives and

verbs, and act as redundant number, gender and person markers. The presence or absence of certain vowels in the desinence is the deciding factor metaphonic alternations. Accentual alternations affect the vowels [a] and [o] in verb forms and are dependent upon stress.

Examples of metaphonic vowel alternations:

- a) [e] becomes [ɛa] when followed by a syllable containing [ə].

<u>creț</u>	[krets]	'curly m sg'
<u>creață</u>	[kreátsə]	'curly f sg'

- b) [ea] becomes [e] when followed by a syllable containing [e], [ury] or a form of word-final i.,

<u>cîntăreață</u>	[kɨntərəátsə]	'singer f'
<u>cîntărete</u>	[kɨntərə́tse]	'singers f'
<u>ceață</u>	[čátsə]	'fog'
<u>cețuri</u>	[čétsury]	'fogs'
<u>dimineață</u>	[dimineátsə]	'morning'
<u>dimineți</u>	[diminétsy]	'mornings'

- c) [e] becomes [a] when followed by a syllable containing [ə].

<u>iert</u>	[yert]	'I excuse'
<u>iartă</u>	[yártə]	'(s)he excuses'

- d) [a] becomes [e] when followed by a syllable containing [e], [ury] or a form of word-final i.

<u>masă</u>	[másə]	'table'
-------------	--------	---------

<u>mese</u>	[mése]	'tables'
<u>iarbă</u>	[yárbə]	'grass'
<u>ierburi</u>	[yérbury]	'grasses'
<u>viață</u>	[vyátsə]	'life'
<u>vieți</u>	[vyétsy]	'lives'

e) [o] becomes [ɔa] before a syllable containing [e] or [ə].

<u>pot</u>	[pót]	'I can'
<u>poate</u>	[pɔáte]	'(s)he can'
<u>rog</u>	[róg]	'I ask'
<u>roagă</u>	[rɔágə]	'(s)he asks'

f) [ɔa] becomes [o] before a syllable containing a form of word-final i.

<u>floare</u>	[flɔáre]	'flower'
<u>flori</u>	[flóry]	'flowers'

g) [ə] becomes [e] before a syllable containing [e] or a form of word-final i

<u>măr</u>	[mér]	'apple'
<u>mere</u>	[mére]	'apples'
<u>văr</u>	[vér]	'cousin m'
<u>veri</u>	[véry]	'cousins m'

h) [a] becomes [ə] before a syllable containing [ury] or a form of word-final i

<u>lampă</u>	[lámpə]	'lamp'
<u>lămpi</u>	[ləmpy]	'lamps'
<u>carne</u>	[kárne]	'meat'

<u>cărnuri</u>	[kə'rnury]	'meats'
----------------	------------	---------

Examples of accentual vowel alternations:

Examples:

i) [a] becomes [ə] when unstressed

<u>cad</u>	[kád]	'I fall'
<u>cazi</u>	[kázy]	'you fall'
<u>cade</u>	[káde]	'she falls'
<u>cădem</u>	[kədém]	'we fall'
<u>cădeți</u>	[kədétsy]	'you fall'
<u>cad</u>	[kád]	'they fall'

j) [o] becomes [u] when unstressed

<u>rog</u>	[róg]	'I beg'
<u>rogi</u>	[rój]	'you beg'
<u>roagă</u>	[roágə]	'he begs'
<u>rugam</u>	[rugám]	'we beg'
<u>rugati</u>	[rugátsy]	'you beg'
<u>roagă</u>	[roágə]	'they beg'

3.4 Consonants

There are 20 consonant phonemes in Romanian:

	p	b	m	f	v	t	d	n	l	r
syllabic	-	-	-	-	-	-	-	-	-	-
sonorant	-	-	+	-	-	-	-	+	+	+
consonantal	+	+	+	+	+	+	+	+	+	+
anterior	+	+	+	+	+	+	+	+	+	+
coronal	-	-	-	-	-	+	+	+	+	+
high	-	-	-	-	-	-	-	-	-	-
low	-	-	-	-	-	-	-	-	-	-
back	-	-	-	-	-	-	-	-	-	-
continuant	-	-	-	+	+	-	-	-	+	-
strident	-	-	-	+	+	-	-	-	-	-
delayed release	-	-	+	+	+	-	-	+	+	-
voiced	-	+	+	-	+	-	+	+	+	-
nasal	-	-	-	-	-	-	-	+	-	-
lateral	-	-	-	-	-	-	-	-	+	-

	ts	s	z	č	ȝ	š	ž	k	g	h
syllabic	-	-	-	-	-	-	-	-	-	-
sonorant	-	-	-	-	-	-	-	-	-	-
consonantal	+	+	+	+	+	+	+	+	+	+
anterior	+	+	+	-	-	-	-	-	-	-
coronal	+	+	+	+	+	+	+	-	-	-
high	-	-	-	+	+	+	+	+	+	-
low	-	-	-	-	-	-	-	-	-	+
back	-	-	-	-	-	-	-	+	+	-
continuant	-	+	+	-	-	+	+	-	-	+
strident	+	+	+	+	+	+	+	-	-	-
delayed release	+	+	+	+	+	+	+	-	-	+
voiced	-	-	+	-	+	-	+	-	+	-
nasal	-	-	-	-	-	-	-	-	-	-
lateral	-	-	-	-	-	-	-	-	-	-

There are no geminate consonants in Romanian except for a few prefixed forms like innascut (cf. Sp. innato), which are all neologisms and therefore were not affected by some (or any) of the historical phonological changes.

An important variant of [n] is [ɲ], which always occurs before velar consonants within words and in normal

speech even across word boundaries.

The sounds [k] and [g] are always palatalized ([ky] and [gy]) before the palatal vowels [e] and [i].

3.5 Consonantal Alternations

The alternations [k/č], [g/ǵ], [t/ts], [d/z], [s/š], and [z/ž] as outlined in 2.2 depend on the presence or absence of a form of unstressed, word-final i as the following segment. Other alternations include the following:

a) [k] becomes [č] before [e]

<u>plec</u>	[plek]	'I leave'
<u>plece</u>	[pléče]	'(s)he leave subj'

b) [g] becomes [ǵ] before [e]

<u>aleg</u>	[aleg]	'I choose'
<u>alege</u>	[aléǵe]	'(s)he chooses'

c) [sk] becomes [št] before [e] or a form of i

<u>cresc</u>	[kresk]	'I grow'
<u>creşti</u>	[kreštȳ]	'you grow'
<u>creşte</u>	[kréšte]	'(s)he grows'
<u>creştem</u>	[kréštem]	'we grow'
<u>creşteţi</u>	[kréštetsȳ]	'you grow pl'
<u>cresc</u>	[kresk]	'they grow'
<u>gîscă</u>	[gǵískə]	'goose'
<u>gîşte</u>	[gǵişte]	'geese'

ceașcă [čáškə] 'cup'

cești [češty] 'cups'

d) [s] becomes [š] before [t] or [tr] plus a form of i

trist [trist] 'sad m sg'

triști [trišty] 'sad m pl'

albastru [albástru] 'blue m sg'

albaștri [albáštri] 'blue m pl'

e) [l] and [n] are omitted when preceded by a vowel and followed by a form of word-final i

cal [kal] 'horse'

cai [kay] 'horses'

spun [spun] 'I say'

spui [spuy] 'you say sg'

spune [spúne] '(s)he says'

spunem [spúnem] 'we say'

spuneți [spúnetsy] 'you say pl'

spun [spun] 'they say'

3.6 Stress

Stress in Romanian, as in Latin, is expiratory. Most words are stressed on the penultimate syllable. Notable exceptions are infinitives ending in stressed -a, -ea, -i and -î, which were originally stressed on the penult before the final syllable was lost, all forms of the imperfect tense, and masculine nouns and adjectives ending in a consonant, which lost their final -u late in the development of

Romanian. All of the above are stressed on the final syllable.

Examples:

<u>cîntare</u>	[kɨntáre]	'singing'
<u>a cînta</u>	[a kɨntá]	'to sing'
<u>cînta</u>	[kɨntá]	'(s)he sang imperf'
<u>avere</u>	[avére]	'fortune; estate'
<u>a avea</u>	[a aveá]	'to have'
<u>avea</u>	[aveá]	'(s)he had imperf'
<u>venire</u>	[veníre]	'coming; arrival'
<u>a veni</u>	[a vení]	'to come'
<u>venea</u>	[veneá]	'(s)he came imperf'
<u>coborîre</u>	[koborîre]	'descent'
<u>a coborî</u>	[a koborî]	'to descend'
<u>cobora</u>	[koborá]	'(s)he descended'
CL BARBĂȚUM	[barbá:tum]	'bearded one'
DR <u>bărbat</u>	[bərbat]	'man'

3.7 Modern Romanian Orthography

Until the mid-19th century, Romanian was written with a Cyrillic alphabet. The present Latin-based orthographic system is sometimes phonetic, sometimes phonemic, and as faithful to the Latin or Romance etymon as possible.

The vowel letters a, e, o, u, and non-final i are pronounced as are the corresponding symbols of the

International Phonetic Alphabet (IPA). The letters e, i, o, and u can serve as glides in diphthongs and triphthongs, in which the most open vowel receives the stress. The circumflex-bearing letters â and î both represent the sound [ɛ̃]. The breve-bearing ă represents [ə]. In words of more than one syllable, word-final -i is usually non-syllabic, often devoiced, sometimes mute as explained in 2.2. Word-final -ii is pronounced [i].

The consonant letters b, d, f, h, l, m, p, r, s, t, v, z are pronounced as are the corresponding symbols of the IPA. Hook-bearing ṭ and ṣ are pronounced [ts] and [ʃ], respectively. The letter j represents the sound [ʝ] as in French. The letter n becomes [ŋ] before a velar sound. The letter x is pronounced [ks] in some words and [gz] in others. The letter c represents the sound [k] except before e and i where it is pronounced [ç] as in Italian. The digraph ch appears only before e and i and represents the sound [ky]. Similarly, the letter g represents the sound [g] except before e and i where it is pronounced [j] as in Italian. The digraph gh appears only before e and i and represents the sound [gy]. When unstressed ce, ci, ge and gi precede a non-palatal vowel, the e and i serve only to indicate that the c and g are pronounced [ç] and [j] respectively and are themselves mute.

CHAPTER 4

THE EARLY CHANGES FROM CLASSICAL LATIN TO THE SLAVIC INVASIONS (FIRST THROUGH SIXTH CENTURIES A.D.)

4.1 The Term 'Balkan Romance'

During the first five or six hundred years of our era, Latin phonology underwent many changes, most of which are common to the histories of most or all of the Romance languages. Our ultimate concern here, however, is the development of post-classical Latin in a specific place: the inland eastern provinces of the Roman Empire. Our goal in this chapter is to hypothesize rules and, where necessary and possible, the order in which they applied until the time when one could no longer speak of this popular language as Latin but rather as the beginnings of a new language in its own right. This early era in the history of the Romanian language has traditionally been designated the Balkan Romance period.

4.2 Rules

With the following descriptive phonological rules one can derive sixth-century Balkan Romance from Latin. The parenthetical references, e.g. (P1), are to Otero's 'procesos fonológicos' contained in Otero 1971 and 1976, on

which the rules in this chapter are based. Again, DR refers to modern standard Daco-Romànian or 'limba literară'.

The commentaries of researchers such as Väänänen, Leumann, Tekavčić, Rohlfs, Vasiliu, and Sala, especially those containing references to graffiti and other epigraphical remains, are cited as evidence that a change was underway.

The symbol >> indicates that intermediate steps have been omitted. The present DR (Daco-Romanian) form is given after the symbol >> for purposes of comparison.

A synoptic list of all the rules may be found at the beginning of Chapter 8.

R1 Loss of [h]

Formally,

h > Ø

Examples:

hómo > ómo >> DR om 'man'

prehéndere > preéndere >> DR a prinde 'to grasp'

kohórtem > koórtem >> DR curte 'courtyard'

habé:re > abé:re >> DR a avea 'to have'

habé:bat > abé:bat >> DR avea '(s)he had imperf'

hibérna > ibérna >> DR iarnă 'winter'

According to Väänänen, intervocalic [h] began to disappear in the pre-literary period. At first the omission

of [h] was considered a vulgar rusticism. Many examples of its omission and hypercorrect inclusion are found in the Pompeian inscriptions and elsewhere. There is no trace of this original CL [h] in any of the Romance languages. The [h] in modern Romanian is from Slavic, Turkish, Hungarian and German loan words. See Leumann (1977:173).

R2A Monophthongization of AE [ay] (P3A)

Formally,

ay > ε:

Examples:

káylum > kɛ:lum >> DR cer 'sky'

dáydam > dɛ:dam >> DR zadă 'larch tree' (var. of TAEDA)

fáynum > fɛ:num >> DR fîn 'hay'

According to Väänänen, the monophthongization of AE took place before that of OE, so it seems logical to order the rules accordingly (Väänänen 1974:95).

The monophthongization of [ay] resulted in the creation of a vowel sound unique in the CL scheme in that it was the only lax vowel and the only long vowel that did not have a short counterpart.

R2B Monophthongization of OE [oy]

Formally,

oy > e:

Examples:

fóytus > fé:tus >> DR făt 'fetus'

fóyta > fé:ta >> DR fată 'girl'

The Roman scholar and encyclopedist Varro writes that this change began in rustic speech (Väänänen 1974:94). As a result of R2B, [oy] merged with [e:].

Of CL's three diphthongs, [aw] was the most resistant to monophthongization. In post-classical Latin, [o] was considered a rustic and vulgar variant of [aw]. Though the great majority of Balkan Romance words containing [aw] did not undergo this change, a few words in Romanian may have developed from the rustic variant:

awríkulam > o:ríkulam >> DR ureche 'ear'

káwda > kó:da >> DR coadă 'tail'

But normally,

áwrum >> DR aur 'gold'

táwrum >> DR taur 'bull'

awdí:re >> DR a auzi 'to hear'

In a few words in which there was an [u] in the following syllable [aw] > [a] (Väänänen 1974:97):

awskulta:re >> DR a askulta 'to listen'

R3A Reduction of final [m] (P6A)

Formally,

m > n / ____ # in monsyllabic words

m > ∅ / ____ # in all others

The sporadic loss of word-final [m] is amply attested

beginning in the third century B. C. (Väänänen 1974:136).

Examples:

kwem > kwen >> DR cine 'who(m)'

and

wétulum > wétulu >> DR vechi 'old'

kum >> DR cu 'with'

kwam > kwa >> DR ca 'as'

See also Tekavčić (1972:197-8).³

R3B Reduction of [n] + fricative (P6B)

When [n] is followed by a fricative, the nasal disappears, and the preceding vowel becomes long.

Formally,

$$V \quad n \quad \begin{bmatrix} -\text{son} \\ +\text{cont} \end{bmatrix} > V \quad \emptyset \quad \begin{bmatrix} -\text{son} \\ +\text{cont} \end{bmatrix}$$

[+long]

Examples:

ménsa > mé:sa >> DR masă 'table'

pénsu > pé:su >> DR păs 'grievance'

dénsu > dé:su >> DR des 'often'

mensúra > me:súra >> DR măsură 'measure'

konsúere > ko:súere >> DR a coase 'to sew'

The deletion of [n] before [s], the first attestation of which occurs in the third century B.C., is one of the most constant traits of "vulgar" texts (Väänänen 1974:132).

Cf. Apppendix Probi: mensa non mesa.

Some past participles are exceptions to this rule:

prénsu >> DR prins 'caught pp'

respónsu >> DR răspuns 'answered pp'

R4A Syncope (P4)

A high vowel is deleted in an internal syllable that is usually post-tonic when this vowel is preceded by a single resonant and followed by a single consonant or vice versa.⁴

Formally,

$$\begin{bmatrix} V \\ +\text{high} \\ -\text{long} \end{bmatrix} > \emptyset / V C_i \text{ — } C_j V$$

where either or both C_i and $C_j = R$

This syncope seems not to have applied universally in Romanian. One must verify the DR form to see if this rule applies.

Examples:

oríkula > oríkla >> DR ureche 'ear'

ókulu > óklu >> DR ochi 'eye'

wétulu > wétlu >> DR veche 'old'

ássula > ásla >> DR aschie 'splinter'

stábulu > stáblu >> DR staul 'stable'

weterá:nu > wetrá:nu >> DR bătrîn 'old man'

kálidu > káldu >> DR cald 'hot'

dóminu > dómnu >> DR domn 'lord'

dómina > dómna >> DR doamnă 'lady'

wíride > wírde >> DR verde 'green'

There are many exceptions, for example

lépore remains lépore >> DR iepure 'hare'

máskulu remains máskulu >> DR mascur 'castrated pig'.

Cf. Appendix Probi: oculus non oclus, viridis non virdis, speculum non speclum, auricula non oricla.

Väänänen (1974:99) seems to see syncope as a recurrent rule. "From a chronological viewpoint, cases of syncope form in echelon along the different stages of Latin from the preliterate era through the Romance phase."

The syncopes that occurred throughout Latin's history are characterized by the persistence, for an indeterminate amount of time, of the non-syncopated form alongside the syncopated form, the former generally considered to be the more correct. The eastern Romance languages seem to be more conservative:

fráksinu >> DR frasin, It. frassino, Sp. fresno 'ash tree'

hedera >> DR iederă, It. edera, Sp. hiedra

It is logical to conclude that the syncopes which affected just one part of the Romance speaking world are more recent than those which affected all Romances.

R4B Dissimilation of [tɫ]: [tɫ] > [kɫ]

Formally:

t > k / ____ l

Since the combination [tɫ] seems to have been phonologically unacceptable the change [tɫ] > [kɫ] must have

taken place concurrently with or immediately after R4A.

According to Väänänen, the combination [sl] needed a 'consonant of passage', namely [k], before which the [ss] naturally degeminated (Väänänen 1974:135).

Examples:

wétlu > wéklu >> DR veche 'old'

ássla > áskla >> DR aschie 'splinter'

R5A Deletion of word-final oral or nasal stop (P7)

Formally,

$\begin{bmatrix} C \\ -cont \end{bmatrix} > \emptyset / ___\#$

Examples:

káput > kápu >> DR cap 'head'

ténet > téne >> DR ține '(s)he holds'

widet > wíde >> DR vede '(s)he sees'

rí:det > rí:de >> DR rîde '(s)he laughs'

habé:bat > abé:ba >> DR avea '(s)he had imperf'

póst > pós >> DR apoi 'afterwards'

sí:k > sí: >> DR și 'and'

du:k > du: >> DR du 'lead imperat'

nómen > nóme >> DR nume 'name'

sángwen > sângwe >> DR sînge 'blood'

non > no >> DR nu 'no; not'

In CL, words could end in only nine consonants: [b,d,k,l,m,n,r,s,t]. Of these only seven were ever phrase final:

[k,l,m,n,r,s,t]. Final [m] fell in R3A, final [k,n,t] fell in R5. Final [s] seems to have been retained till the very end of the Post-Classical period (see R25).

R5B Addition of word-final [e] after [l]

Formally,

$\emptyset > e / l ___ \#$

Examples:

mel > méle >> DR miere 'honey'

fel > féle >> DR fiere 'bile'

sál > sále >> DR sare 'salt'

Tekavčić (1972:208) suggests that CL words ending in [l] received a paragogic final [e] before or the onset of the Common Romanian period (6th century). At some point, kuen (< QUEM) also received a final -e:

kwem > kwen > kwene >> DR cine 'who(m)'⁵

R5C Metathesis of word-final [r]

Formally,

V	r	#	>	r	V	#
1	2	3		2	1	3

Examples:

kuátt(w)or > kwáttro >> DR patru 'four'

súper > súpre >> DR spre 'towards'

ínter > íntre >> DR între 'between'

or the final [r] was simply eliminated:

fráter >> DR frate 'brother'

per >> DR pe 'on'

R6 Fronting of [w] to [β] (P10)

Formally,

w > β

Examples:

wení:re > βení:re >> DR a veni 'to come'

wíde > βíde >> DR vede '(s)he sees'

wé:na > Bé:na >> DR víně 'vein'

wéklu > Béklu >> DR vechi 'old'

wetrá:nu > βetrá:nu >> DR bătrîn 'old person'

wákka > βákka >> DR vacă 'cow'

egzwolá:re > egzβolá:re >> DR a zbura 'to fly'

wéntu > Béntu >> DR vînt 'wind'

wí:nu > βí:nu >> DR vin 'wine'

ó:wu > ó:βu >> DR ou 'egg'

nówu > nóβu >> DR nou 'new m sg'

kérwu > kérβu >> DR cerb 'stag'

serwá:re > serβá:re >> DR a serba 'to celebrate'

alwí:na > alβí:na >> DR albină 'bee'

prima wéra > prima Béra >> DR primăvară 'spring'

ad de wéro > ad de Béro >> DR adevăr 'truth'

According to Väänänen (1974:112), beginning in the 1st century AD, Latin V is frequently transcribed in Greek with the letter β, which at that time was most likely the

bilabial continuant [β].⁶

As a result of R6, [b] and [β] existed in Latin. When [b] and [β] were merely positional variants as in modern Spanish, there would not have been any great problem keeping them apart since the environment determines the use of one or the other. After R6, however, they are no longer in complementary distribution and are phonetically so close that there may have been a tendency to confuse them.

At first glance, it seems that R21 [b] > [β] ought to apply in conjunction with R6, and it is possible that the tendency for at least medial [b] to become a fricative began at this time. However, a few rules which must for various reasons be ordered later indicate that R21 did not apply universally until after they applied. See the commentaries to R21 for details.

R7 Fortition of syllable initial [y]: [y] > [dy]

Formally,

y > dy / \$ ____

Examples:

yurá:re > dyurá:re >> DR a jura 'to swear'

yoká:re > dyoká:re >> DR a juca 'to play'

yó:wis > dyó:wis >> DR joi 'Thursday'

The fortition of [y] probably began in intervocalic position in which it had always been geminated (Väänänen

1974:115). In R18, [dy] from this and other sources becomes [j].

R8 Simplification of K[s]K,K[z]K cluster (P8)

Formally,

[-son] > Ø / ____ [+strid] [-son]

Examples:

abskóndere > askóndere >> DR a ascunde 'to hide'

egzβolá:re > ezβolá:re >> DR a zbura 'to fly'

ekspó:nere > espó:nere >> DR a spune 'to say'

For Balkan Romance, R6 (P10) must apply before R8 (P8) to include changes such as [egzβolá:re] > [ezβolá:re].

R9 Deletion of [k]: [nkt] > [nt]

Formally,

k > Ø / n ____ t

Examples:

sánkту > sántu >> DR sfînt 'holy; saint'

únktu > úntu >> DR unt 'grease, butter'

frá(n)ktu > frántu >> DR frîmt 'broken'

Note that Otero's P12 ([kt] > [xt]) does not apply in Balkan Romance. According to Otero (1976:40), P12 "es el proceso que escinde el romance occidental de los dialectos del centro y sur de Italia y de los Balcanes."

R10A Assimilation of [r] to [s] (sporadic) (P13A)

Formally,

r > s / ____ s

Examples:

dórsu > dóssu >> DR dos 'backside'

deórsu > deóssu >> DR jos 'down'

sú:rsu > sú:ssu >> DR sus 'up'

but

úrsu remains úrsu >> DR urs 'bear'

persika remains persika >> DR piersică 'peach'

R10B Assimilation of [p] to [s] (P13)

Formally,

p > s / ____ s

Examples:

skrípsi > skríssi >> DR scrise '(s)he wrote'

Note that Romanian conserves CL [pt]:

septe >> DR șapte 'seven'

septimána >> DR săptămână 'week'

R10Ca Partial assimilation of velars (P12, P13)

After a stressed vowel a velar stop, oral or nasal,
becomes bilabial before an alveolar consonant:

[ks] > [ps], [kt] > [pt], [ɣn] > [mn]

Formally,

$$\begin{bmatrix} -\text{cont} \\ -\text{ant} \\ -\text{cor} \end{bmatrix} > [+ant] / \begin{bmatrix} \text{V} \\ +\text{acc} \end{bmatrix} \text{ --- } \begin{bmatrix} \text{C} \\ +\text{ant} \\ +\text{cor} \end{bmatrix}$$

Examples:

kóksa > kópsa >> DR coapsă 'thigh'
fríksi > frípsi >> DR fripse '(s)he fried'
lákte > lápte = DR lapte 'milk'
fáktu > fáptu >> DR fapt 'done pp'
óкто: > ópto: >> DR opt 'eight'
intelléktu > intelléptu >> DR înțelept 'wise'
diréktu > diréptu >> DR drept 'direct'
laktúka > laptúka >> DR lăptucă 'lettuce'
lúkto > lúpto >> DR lupt 'I struggle'
nókte > nópte >> DR noapte 'night'
péktine > péptine >> DR pieptene 'comb'
péktu >> DR piept 'chest'
díɣnu > dímnu >> DR demn 'worthy'
líɣnu > límnu >> DR lemn 'wood'
púɣnu > púmnu >> DR pumn 'fist'
síɣnu > símnu >> DR semn 'sign'

R10Cb Total assimilation of velars

Before stress, total assimilation seems to have occurred in most cases:

Formally,

$$\begin{bmatrix} -\text{cont} \\ -\text{ant} \\ -\text{cor} \end{bmatrix} > \begin{bmatrix} +\text{ant} \\ +\text{cor} \end{bmatrix} / \begin{bmatrix} \text{v} \\ -\text{ac} \end{bmatrix} \text{ --- } \begin{bmatrix} \text{c} \\ +\text{ant} \\ +\text{cor} \end{bmatrix}$$

Examples:

laksá:re > lassá:re >> DR a lasa 'to let'

eksí:re > essí:re >> DR a ieși 'to go out'

fraksinétu > frassinétu >> DR frășinet 'ash wood'

ko:ɲnó:skere > ko:nnó:skere >> DR a cunoaște 'to know'

but

koɲnátu >> DR cumnat 'brother-in-law'

With R10C we posit, with Vasiliu, that Latin -GN- became Romanian [mn] in the same way that [kt] > [pt] and [ks] > [ps].

It is generally held and well supported that Latin [g] > [ɲ] before [n].⁷ In Sardinian, for example, it is reasonably certain that the [nn] developed from [ɲn] e.g., *agnile 'sheepcot' > annile, LIGNA (pl) > linna 'firewood' (coll.), MAGNU > mannu, PIGNU > pinnu, SIGNU > sinnu, etc. It is equally well supported, however, that most Romance reflexes of Latin etyma containing the sequence -GN- did not develop from [ɲn] but rather from [gn], the latter perhaps being a restored spelling pronunciation.

The following derivation illustrates two routes suggested for the development of Romanian [mn]:

	Sard	Rom A	It/Sp/Pt	Rom B
CL	líɣna-	líɣnu-	lígnu-	lígnu-
Assimil	línna	-----	-----	-----
R10Cb	-----	límnu	-----	-----
[g]>[+cont]	-----	-----	líɣnu	líɣnu
V>[+low]	-----	lémnu	léɣno	léɣnu
[g]> glide	-----	-----	léyno	léwnu
[n]>[-ant]	-----	-----	leño	-----
[w]>[v]	-----	-----	----	lévnu
Assimil	-----	-----	----	lémnu
[u]>ø/_#	-----	lemn	----	lemn

Using the simplicity criterion it seems better to opt for Rom A thereby including nasals in R10C, especially since Rom B would require the formulation of additional rules specific to this phenomenon. For those who hold that there is a close historical relationship between Italian dialects and Romanian, Rom B may seem preferable.⁸

R11 Labialization of [kw] and [gw] before [a]

[kwa] > [pwa] and [gwa] > [bwa]

Formally,

$$\begin{bmatrix} -\text{son} \\ -\text{ant} \\ -\text{cor} \end{bmatrix} > \begin{bmatrix} +\text{ant} \end{bmatrix} / \text{___ wa}$$

Examples:

ákwa > ápwa >> DR apă 'water'

ékwa > épwa >> DR iapă 'mare'

kwáttro > pwáttro >> DR patru 'four'

kwadragésima >> DR păresimi 'lent'

língua > límbwa >> DR limbă 'tongue'

interrogá:re >> intregwáre > intrebwáre >> DR întrebare
'to ask'

Vasiliu sees this is an assimilatory rule (1968:92).

Sometime before R11 and in anticipation of R17 (w > Ø),
QUALIS, QUAM, QUANDO, QUANTUS, SQUAMA, etc., must have lost
the wau becoming [kále], [ka], [kándó], [kántu], [skáma],
forms to which R11 could not apply.

CL QU and GU both became [b(b)] in Sardinian but in
different environments and presumably in a different manner
than in Romanian.⁹

R12 Prothesis: initial [s]C > [is]C (P14)

Formally,

$$\emptyset > \begin{bmatrix} V \\ +\text{high} \\ -\text{back} \end{bmatrix} / \# \text{ — } s \begin{bmatrix} C \\ -\text{son} \end{bmatrix}$$

Examples:

skála > iskála >> DR scară 'ladder; staircase'

stá:re > istá:re >> DR a sta 'to stand; to stay'

skáma > iskáma >> DR scamă 'lint'

spáta > ispáta >> DR spată 'sword; shoulder blade'

According to Väänänen (1974:108), from the 2nd century onward, epigraphic remains show that prothesis, instead of obeying a euphonic principle (as in Italian), occurred word initially regardless of whether the preceding word ended with a vowel or a consonant. Later, texts from Gaul, where final consonants were more stable than in the East, appeared to favor prothesis after a word ending in a consonant, which seems to indicate a tendency to avoid difficult consonant clusters. In the East, however, where practically all final consonants had been lost (R5), a later apheresis (R14) removed the epenthetic vowel.

R13A Lengthening of second vowel in vowel combinations
and concomitant shift in stress (P15A)

Formally,

$$\begin{array}{ccccc} \left[\begin{array}{c} V \\ -\text{low} \\ +\text{acc} \end{array} \right] & V & > & \left[\begin{array}{c} V \\ -\text{low} \\ -\text{acc} \end{array} \right] & \left[\begin{array}{c} V \\ +\text{long} \\ +\text{acc} \end{array} \right] \\ 1 & 2 & & 1 & 2 \end{array}$$

Examples:

muliere > mulié:re >> DR muiere 'woman'

pariete > parié:te >> DR perete 'wall'

ariete > arié:te >> DR arete 'ram' (regional)

kapreolu > kapreó:lu >> DR căprior 'roebuck'

With R13A, stress shifted to the new long vowel in accordance with Latin stress rules, which were still in

effect.

R13B Raising of [o] and [e] (P15B)

Short, unstressed [o] and [e] become [u] and [i] respectively when adjacent to another vowel.

Formally,

$$\begin{bmatrix} V \\ -\text{low} \\ -\text{long} \\ -\text{acc} \end{bmatrix} > [+high] \quad // \quad V$$

Examples:

púteu > pútiu >> DR puț 'well'

pálea > pália >> DR paie 'straw'

deóssu > dióssu >> DR jos 'down'

kapreó:lu > kaprió:lu >> DR căprior 'roebuck'

wí:nea > wí:nia >> DR vie 'vineyard'

ká:seu > ká:siu >> DR caș 'cheese'

lákweu > lákwiu >> DR laț 'lasso'

This phenomenon is attested as far back as the CL period (Väänänen 1974:105).

R13C Contraction of Vowel Combinations (P18B)

Two sequential identical vowels coalesce.

Formally,

$$V_i V_i > V_i$$

Example:

mórtuu > mórtu

R13D Loss of syllabicity of high vowels (P15)

The short, unstressed high vowels [i] and [u] become the glides [y] and [w] when adjacent to another vowel.

Formally,

$$\begin{bmatrix} \text{V} \\ +\text{high} \\ -\text{long} \\ -\text{acc} \end{bmatrix} > [-\text{syll}] // \text{V}$$

Examples:

pútiu > pútyu >> DR puț 'well'

kamísia > kamísyu >> DR cămașă 'shirt'

glákie > glákye >> DR gheață 'ice'

Bínia > Bínyu >> DR vie 'vineyard'

ká:siu > ká:syu >> DR caș 'cheese'

This rule seems to be recurrent, i.e. it applies from this point on whenever the environment presents itself.

R13E Laxing of stressed [e] to [ɛ] (P15Da) (Otero 1976:19)

Formally,

$$\begin{bmatrix} \text{V} \\ -\text{long} \\ -\text{high} \\ -\text{back} \\ +\text{acc} \end{bmatrix} > [-\text{tense}]$$

Examples:

lépore > l'épore >> DR iepure 'hare'

véntu > v'éntu >> DR vînt 'wind'

béne > b'éne >> DR bine 'well'

férru > f'érru >> DR fier 'iron'

méle > mǎle >> DR miere 'honey'

This rule results in a split [e]/[ɛ] dependent upon stress. For details concerning the development of [ɛ] and its subsequent diphthongization (R27), see Otero 1985.

R13F Lengthening of word-final [-i(s)]

Formally,

$$\left[\begin{array}{c} \text{v} \\ +\text{high} \\ -\text{back} \end{array} \right] > [+long] / \text{ ______ } (C) \#$$

Example:

fákitis > fákiti:s >> DR faceți

This rule may be necessary to prevent second person plural verb forms from ultimately ending in -te as they do in Italian. In this case R13F would bleed R13G.

According to Bynon (1977:137), Kiparsky has drawn attention to the fact that systematic restructuring may include the conservation of a segment which serves to mark a grammatical category in defiance of a given phonetically motivated rule. Morphological regularity would then be achieved by blocking a phonetically motivated rule in a specific grammatical environment (Bynon 1977:137). If such were the case here, R13G would not apply.

R13G Lowering of [i]: Stressed or unstressed [i] > [e]

(P15Db):

v -long +high -back	>	[-high]
------------------------------	---	---------

Examples:

kamísyā > kamésya >> DR cămaşă 'shirt'

pílu > pélu >> DR păr 'hair'

βíde > Béde >> DR vede '(s)he sees'

píru > péru >> DR păr 'pear tree'

límbwa > lémbwa >> DR limbă 'tongue'

The letter e appears instead of i in Pompeii in stressed and unstressed position. The same occurs in various "vulgar texts" especially towards the end of the VL period (Väänänen 1974:92).

The parallel lowering of [u] > [o], which according to most sources did not occur in the evolution of Romanian, is found very rarely even towards the end of the post-classical period and is evidently subsequent to the change i > e (Väänänen 1974:92).

Still problematic are the DR words

toamnă << tomna << atumna << CL AUTUMN- 'autumn'

cot << CL CUBITUM 'elbow'

roib << CL RUBEUS 'reddish'

which may indicate 1) that R13G also included the universal change [u] > [o], as it did in the western Romances, most

instances of which were changed back to [u] by subsequent rules, 2) that the change [u] > [o] began in BR but died out before it could apply universally, 3) that these words came into BR from the West after [u] > [o] had applied universally there.

R14 Apheresis: Loss of unstressed initial [e] before [s] or [z] before a consonant

Formally,

$$e > \emptyset / \text{---} s \left[\begin{array}{l} +\text{cont} \\ +\text{ant} \\ +\text{cor} \\ +\text{str} \end{array} \right]$$

Examples:

eskála > skála >> DR scară 'stair'

espéro > spéro >> DR sper 'I hope'

espónere > spónere >> DR a spune 'to say'

ezBolá:re > zBolá:re >> DR a zbura 'to fly'

estrapúngere > strapúngere >> DR a străpunge 'to pierce'

This rule erases the effect of prothesis (R12) as well as the initial vowel of words originally beginning with INS-, HIS-, AES- and EX-.

R15 Contraction of vowels (P18Aab)

Formally,

$$\begin{bmatrix} \text{v} \\ \text{-long} \\ \alpha\text{back} \end{bmatrix} > \emptyset / \text{c} \text{ --- } \begin{bmatrix} \text{v} \\ \text{+long} \\ \text{+acc} \\ \alpha\text{back} \end{bmatrix}$$

Examples:

oó: > ó: koórte > kó:rte

eé: > é: preé:ndere > pré:ndere

ié: > é: kwié:tu > kwé:tu

According to Vaananen (1974:104), two consecutive vowels in hiatus having the same or very similar heights tend to contract into one long vowel of the height of one or both of the vowels in question. The results of the application of this rule seem to be attested in Pompeii.

As regards koórte > kó:rte, Ernout and Meillet (131a) mention that the earliest ancient Greek transcriptions show two vowels (as yet unmerged), kóortis (Polybius), but then in Imperial times transcribe as khors/khortes.

R16 Loss of vowel length (P15F)

Formally,

$$\text{v} > [-\text{long}]$$

Examples:

ókto: > ókto >> DR opt 'eight'

kwéro: > kwéro >> DR cer 'I ask'

Bení:re > Beníre >> DR a veni 'to come'

βé:na > Béna >> DR vîná 'vein'

rí:βu > ríβu >> DR rîu 'brook; river'

rí:de > ríde >> DR rîde '(s)he laughs'

zβolá:re > zβoláre >> DR a zburá 'to fly'

abé:ba > abéba >> DR aveá '(s)he had imperf'

βí:nu > βínu >> DR vin 'wine'

This rule, perhaps more than others, indicates the end point of a phonological process which was in progress for centuries. To quote Väänänen (1974:84) (translation mine):

It is evident that the loss of vowel length did not occur suddenly, nor did it occur everywhere at the same moment at all social levels. Judging from the texts, the upsetting of CL rhythm occurred towards the end of the 3rd century. The grammarian Sacerdos, who notes that the loss of distinction in vowel length is a "barbarism of our times", establishes under Diocletian a new 'cursus' or clause, which is no longer founded, like the Ciceronian clauses, on the alternation of long and short syllables, but on a rhythm of accents which do not take into account differences in length (M. Nicolau, *L'origine du cursus rythmique*, Paris 1930, p. 135) One hundred years later Saint Augustine notes (*De Doctr. Christ.* 4, 10, 24) that speakers of Latin in Africa do not distinguish between the short and long vowels in [os] 'mouth' and [o:s] 'bone', for example. The new rhythm is gradually introduced via Christian poetry, although medieval poetry continued, by and large, to observe CL vowel length, especially in stressed syllables. These innovations in rhythm are however only the consequence of a linguistic fact--the loss of opposition of length--realized in the spoken language, and whose beginnings date far back in the past (Cf. G. Straka)."

R17 Deletion of wau before a vowel (P22, T22)

Formally,

w > ∅ / ____ V

Examples:

kwéro > kéro >> DR cer 'I ask'
 kwéne > kéne >> DR cine 'who(m)'
 kwétu > kétu >> DR încet 'slow'
 kwómo > kómo >> DR cum 'how'
 kókwere > kókere >> DR a coace 'to cook'
 sángwe > sánga:>> DR sînge 'blood'
 stîngwere > stîngere >> DR a stînge 'to extinguish'
 úngwere > úngere >> DR a unge 'to grease'
 ápwa > ápa >> DR apă 'water'
 pwáttro > páttro >> DR patru 'four'
 lémbwa > lémba >> DR limbă 'tongue'
 báttwere > băttere >> DR a bate 'to beat'

R18A Gemination of stop when after a vowel and before yod

(P19)

Formally,

V	$\begin{bmatrix} \text{-son} \\ \text{-cont} \end{bmatrix}$	Y	>	V	$\begin{bmatrix} \text{-son} \\ \text{-cont} \end{bmatrix}$	$\begin{bmatrix} \text{-son} \\ \text{-cont} \end{bmatrix}$	Y
1	2	3		1	2	2	3

Examples:

pútyu > púttyu >> DR puț 'well'
 prétyu > préttyu >> DR preț 'price'
 adyutáre > addyutáre >> DR a ajuta 'to help'
 tetyóne > tettyóne >> DR tăciune 'ember'
 brákyu > brákkyyu >> DR braț 'arm'

R18E Assimilation of stops to affricates (P20C)

Formally,

$$C_i C_j > C_j C_j \quad \text{where } C_j \text{ is } \begin{bmatrix} -\text{cont} \\ -\text{ant} \\ +\text{cor} \\ +\text{del rel} \end{bmatrix}$$

Examples:

pútču > púčču >> DR puț 'well'

adjutáre > ajjutáre >> a ajuta 'to help'

brákču > bráčču >> DR braț 'arm'

fákča > fáčča >> DR față 'face'

R19 Depalatalization: [j], [č] > [dz], [ts] after a stressed vowel (P32)

Formally,

$$\begin{bmatrix} -\text{ant} \\ +\text{cor} \\ +\text{del rel} \end{bmatrix} > [+ant] \quad / \quad \begin{matrix} V \\ +\text{acc} \end{matrix} \quad \text{—}$$

Examples:

púčču > púttsu >> DR puț 'well'

bráčču > bráttsu >> DR braț 'arm'

fáčča > fáttsa >> DR față 'face'

pránju > prándzu >> DR prînz 'lunch'

but

ajjutáre remains ajjutáre

teččúne remains teččúne

R20 Attraction of glides (P23)

Formally,

$$\begin{array}{ccccc} & V & Co & & V & Co \\ & +acc & & & +acc & \\ & & \left[\begin{array}{c} +son \\ -cons \\ -syl \end{array} \right] & > & \left[\begin{array}{c} +son \\ -cons \\ -syl \end{array} \right] & \\ 1 & 2 & 3 & & 1 & 3 & 2 \end{array}$$

Examples:

róbbyu > róybbu >> DR roib 'reddish'

ábbya > áybba >> DR aibă '(s)he have subjunc'

skábya > skáybba >> DR zgaibă 'scab'

R21 Spirantization of [b] (P25)

Formally,

$$b > \beta / V \text{ ____}$$

Examples:

abéba > aβéβa >> DR avea '(s)he had imperf'

bébere > béβere >> DR a bea 'to drink'

Word-initial [b] remains unaffected:

béne remains béne >> DR bine 'well'

bráttsu remains bráttsu >> DR 'arm'

R22A Word-initial [β] > [v]

Formally,

$$\beta > v / \# \text{ ____}$$

Examples:

βeníre > veníre >> DR a veni 'to come'

βéntu > véntu >> DR vînt 'wind'

ḡákka > vákka >> DR vacă 'cow'

A few DR words developed [b] in initial position:

ḡerḡék- >> DR berbec 'ram'

ḡetránu >> DR bătrîn 'old man'

These are sporadic results which cannot be accounted for by the phonological rules.

If both CL [b] and CL [w] had merged completely to [ḡ] before R22A, it would be difficult to see why the great majority of CL forms beginning with [w] and those beginning with [b] turn out to have Romanian reflexes in [v] and [b] respectively.

R22B Occlusivization of [ḡ]

[ḡ] becomes a stop after a consonant.

Formally,

[ḡ] > [b] / C ____

Examples:

kérḡu > kərbu >> DR cerb 'stag'

serḡáre > serbáre >> DR a serba 'to celebrate'

alḡína > albína > DR albină 'bee'

zḡoláre > zboláre >> DR a zbura 'to fly'

All of the above forms contain the reflex of CL [w] (and not of CL [b] which presumably never was a fricative in this environment).

R22C [β] becomes the velar glide [w] before a resonant

Formally,

β > w / — [+son
 +cons]

Examples:

stáβlu > stáwlu >> DR staul 'stable'

číβru > číwru >> DR ciur 'sieve' (CL CRIBRUM)

skáβnu > skáwnu >> DR scaun 'chair'

dáβna > dáwna >> DR daună 'damage'

The last two examples are hypothetical forms of CL SCAMNU- and DAMNA-. It is possible that in these forms [m] became [β] via dissimilation sometime before R10C. R22C would then apply to them at this point.

R22D Intervocalic [β] > [g] in the presence of [u]

Formally,

β > g // u (sporadically)

Examples:

úβer > úger >> DR uger 'udder'

néβula > négula >> DR negură 'fog, mist'

rúβu > rúgu >> DR rug 'blackberry'

fáβul- > fágule >> DR fagure 'honeycomb'

néβu > négu >> DR neg 'wart'

The change [β] > [g] may have had an intermediary step [ɣ]. The substitution of [β] for [ɣ] is found today in dialects of Spanish: [aβúxa] instead of [aɣúxa] 'needle'.

[úβer] > [úger] also aids in the ordering of R22D in relation to R24 in which [úger] > [úǵer].

22E [β] > ∅ / elsewhere

Formally,

β > ∅

Examples:

nóβu > nów >> DR nou 'new m sg'

ríβu > ríw >> DR rîu 'river'

aβúnklu > (a)únklu >> DR unchi 'uncle'

béβere > béere >> DR a bea 'to drink'

The root of the DR verb a avea, however, seems to be an exception:

aβére > avére >> DR a avea 'to have'

aβéβa > avéa >> DR avea '(s)he had imperf'

as do the reflexes of some compound words:

prima βéra >> primavera >> DR primăvară 'spring'

ad de βéro >> addevero >> DR adevăr 'truth'

With the application of R6, [w] > [β] in all environments. In R21, [b] > [β] but not word-initially, which creates a merger word-medially, i.e. by R21 both CL [w] and CL [b] have become [β]. In R22, non-initial [β] changes to ∅, [w], [v] or [g] regardless of whether it stemmed originally from CL [w] or early [β]. The above is illustrated in the following derivation:

CL	wení:re	nó:wu	abé:ba:	bónu
R6	Bení:re	nó:βu	-----	----
R21	-----	-----	aβé:βa:	----
R22	veníre	nóu	avéa	----
R32	-----	---	aveá	----
DR	a veni	nou	avea	bun
	'to come' 'new' 'to have' 'good' ¹⁰			

Note at this point that the voicing of [p], [t], [k] after a vowel and before a vowel or resonant (P27 Lenition) did not apply to Eastern Romance, that is, to Romanian or the Central and Southern Italian dialects (Otero 1971:121).

R23 Degemination of consonants except the alveolar resonants [ll], [nn] and [rr] (P28)

Formally,

$$C_i C_i > C_i \quad \text{where } C_i \neq \begin{bmatrix} +\text{ant} \\ +\text{cor} \\ +\text{son} \\ +\text{cons} \end{bmatrix}$$

Examples:

addúko > adúko >> DR aduc 'I bring'

bráttsu > brátsu >> DR braț 'arm'

aǰǰutáre > aǰutáre >> DR a ajuta 'to help'

ǰóssu > ǰósu >> DR jos 'down'

vákka > váka >> DR vacă 'cow'

but

kaállu remains kaállu >> DR cal 'horse'

ánnu remains ánnu >> DR an 'year'

orrésko remains orrésko >> DR urăsc 'I hate'

According to Väänänen (1974:124), the reduction of geminates did not occur until the eve of the fixation of the Romance dialects.

R24 Affrication of velars: [k], [g] > [č], [ǵ] before a front vowel (P21)

Formally,

$$\begin{bmatrix} -\text{cont} \\ -\text{ant} \\ -\text{cor} \\ -\text{del rel} \end{bmatrix} > \begin{bmatrix} +\text{cor} \\ +\text{del rel} \end{bmatrix} > \text{---} \begin{bmatrix} \text{v} \\ -\text{back} \end{bmatrix}$$

Examples:

kélu > čélu >> DR cer 'sky'

kérbu > čérbu >> DR cerb 'stag'

gélu > jélu >> DR ger 'frost'

kéro > čéro >> DR cer 'I ask'

skío > sčío >> DR ştiu 'I know'

skis > sčis >> DR ştii 'you know sg'

R25A Development of final -s in monosyllabic Words (T23)

In monosyllabic words, [s] was somehow replaced with [y]. The formulation [s] > [y] would violate the principle of naturalness. However, the following steps series seems acceptable since each step is natural:

$$[s] > [h] > [ç] > [y] / \text{---} \#$$

Examples:

nos > noh > noç > noy = DR noi 'we'

vos >> voy = DR voi 'you pl'

pos >> poy >> DR apoi 'afterwards, then'

tres >> trey = DR trei 'three'

The rationale for naturalness is as follows: [s] > [h] in syllable-final position in Cuban Spanish and in the evolution of French. [h] and [ç] are interchangeable for some speakers in Spanish words such as 'jinete'. [ç] and [j] differ by just one feature, namely voicing.

As for the loss of final [s] in polysyllabic words, the following phonological series are natural and possible but highly improbable given the time span and are attested nowhere:

is > ih > i

es > eh > e

as > ah > aç > ay > ey > e: > e / ____ #

os > oh > oç > oy > öy > i: > i

[ay] > [ey] in the history of Portuguese: lakte- > laxe > layte > leyte. The [oy] of Classical Greek is pronounced [i] in Modern Greek.

R25B Loss of final [s] in polysyllabic words

can reasonably only be formulated thus:

Final [s] > Ø

Formally,

s > Ø / ____ #

Although graffiti and epigraphical remains may seem to indicate that the accusative noun plurals ending in [s] had completely supplanted the nominative plurals ending in [i] and [e] throughout the Empire, the simplest linguistically based explanation seems to be the following: After R25B applied (to polysyllabic words), there was confusion of singular and plural due to similarity or identity of form. On some level of speech the Latin nominative plural endings [i] and [e] must have still existed in the collective grammar of the speech community of the eastern part of the Roman Empire and were called forth to provide a clear distinction between singular and plural.

Also, since the second person singular marker for most verb forms was [s], and since the loss of [s] usually caused identity of form with the third person singular, the therapy may have been to generalize the ending of the second person singular of the fourth conjugation since it ended in [i] and the corresponding third person singular ended in [e].

Examples of simple loss of final [s]:

áudis > áudi >> DR auzi [aúzy] 'you hear sg'

fáĉetis > fáĉeti >> DR faceți [fácetsy] 'you do pl'

Examples of the restoration of the Latin nominative plural endings:

kásas > kása > káse >> DR case 'houses'

lúpos > lúpo > lúpi >> DR lupi [lúpy] 'wolves'

R25C Generalization of word-final [i] as grammatical marker

Examples of the generalization of the [i] of the second person singular of Latin's fourth conjugation minus its final [s] (audi:s > audis > audi):

kántas > kánta > kánti >> DR cînți 'you sing sg'

vedes >> vede > vedi >> DR vezi 'you see sg'

See Tekavčić's brilliant account of loss of final [s] (1972:199-206).

It is certainly not within the scope of this dissertation to discuss in any more detail much less to solve the highly involved and debated question of the fate of final [s] in Romanian and central and southern Italian. Whatever the phonological development might have been, the result was that adjectives and most nouns may have taken on a regularized morphological shape during the post-classical period when Balkan- and Italo-Romance may still have been in contact.¹¹

CHAPTER 5

THE DEVELOPMENT OF COMMON ROMANIAN (6TH-11TH CENTURIES A.D.)

5.1 The Common Romanian Period

Common Romanian (CR) is the term Coteanu used to designate the stage of the Latin-Romanian continuum in which it is probable that Daco-Romanian and the southern dialects, i.e. Aromanian, Megleno-Romanian and Istro-Romanian, formed a more or less unitary linguistic whole.

Although the dates of both the beginning and the end of this era are disputed, we shall accept Sala's hypothesis that the Common Romanian period began in the 6th century A.D. and ended in the 11th with the geographic and linguistic separation of the Aromanians.

During this period, CR speakers were effectively isolated from the rest of the Romance-speaking world due mainly to the disintegration of the western Roman Empire and its institutions and the subsequent encircling of the Romance speakers by vast numbers of invading Slavs. After mercilessly ruining and pillaging whatever they encountered, the Slavs settled permanently in the new areas.

The Romance-speaking population was not annihilated by the Slavs, nor was its language replaced by Slavic. The

major effect of the Slavic incursions was the destruction of national and linguistic boundaries by means of the dispersal of concentrations of Romance-speakers and their absorption into Slavic-speaking areas.

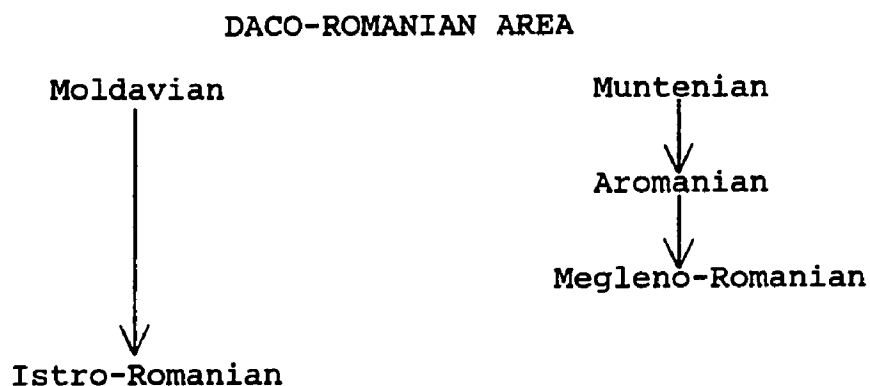
Since historical data on the subject is scant, interpretation and speculation on what ensued has been rampant. Of all the scenarios that have come to my attention, Shevelov's interpretation (1965:159) seems to be the most probable:

If an ethnographic or linguistic map of the Balkans of that time existed, it would not have any uniformly colored large surfaces, but instead its character would be that of a mosaic inlaid with tiny pieces. Slavic-speaking people lived side by side with Romance speaking (as well as the remnants of those who spoke the pre-Romance languages of the area). It took several centuries for centers to crystallize, where one particular language would prevail. If present-day Romania emerged in the Middle Ages as predominantly Romanian in its language, it is not because it had been settled by the Romans alone, but because in the course of mutual Slavic-Roman influences the Roman speech conquered; and if present day Bulgaria emerged as a Slavic speaking country in the Middle Ages, it is not because the Romance speaking population inhabiting the area before the Slavic invasions was expelled or annihilated physically, but because Slavic overcame Romance during a prolonged period of competition. Slavic was absorbed into Romanian in Romania, and Romanian was absorbed into Slavic in Bulgaria.

Since the Slavs were more numerous and powerful, Slavic became the language of administration and trade. And because the Slavs were Christianized by the Eastern Church, the language of cult for all Christians in the area was probably Old Slavic. So it seems obvious that many

inhabitants of the Balkans would have been at least bilingual. As a result, the lexicon of Romanian was greatly enlarged with Slavic words (a parallel might be drawn here between Romanian and its Slavic superstrat and the English of the Middle Ages with its French superstrat). The phonetic inventory of Romanian became enriched with several new sounds, as well. The same Slavic influence is found in all four Romanian dialects.

It is not clear exactly when or how Common Romanian split up into separate dialects. What seems to be confirmed by linguistic evidence is that there were two principal northern dialects, Muntenian and Moldavian, both of which are classified geographically as Daco-Romanian. Aromanian seems to have stemmed from Muntenian. Later, Megleno-Romanian developed from Aromanian. Later still, Istro-Romanian separated from Moldavian, as illustrated below:



Since there are no texts from this period, in order to establish rules for CR we must compare the four major

dialects. Those changes that are common to all four dialects or, at least, to DR and AR might seem to be input for rules for the CR period. Vasiliu, however, rightly cautions that a few changes common to some or all of the dialects may actually be parallel innovations that occurred after the CR period.

5.2 Rules

The following rules will derive Common Romanian from its Balkan Romance antecedent.

R26A [rr] > trill

Formally,

rr > [+rep]

Examples:

orrésko >> DR urăsc 'I hate'

orrétu >> DR urât 'ugly'

We are assuming here that -RR- in Classical Latin was somehow a doubled version of a single flap [r]. In the event that -RR- was already a multiple vibrant in CL, this rule would not be necessary.¹²

R26B Initial [r] > [rr]

Formally,

[r] > [+rep] / # ____

Examples:

re w > rrew >> DR rău 'bad'

rívu > rrívu >> DR riu 'river'

This change is important in the development of Romanian since, in Daco-Romance, [e] became [ə] after [rr] (R50) but not after [r]. [i] also became [ɨ] (R50) in the same environment.

R27A Tensing of [ɛ] before nasal + oral consonant

Formally,

V > [+tense] / ____ N C
[-nas]

Examples:

témpu > témpu >> DR timp 'time'

dénte > dénte >> DR dinte 'tooth'

lémba > lémba >> DR limbă 'tongue; language'

This rule is necessary to bleed R27B. If [ɛ] had diphthongized in these cases, the presence of yod would have led to the palatalization of consonants preceding yod, as illustrated in the following derivation:

	témpu	dénte	lémba
R27B	*tyémpu	*dyénte	*lyémba
R41	-----	-----	*λémba
R42	*tsémpu	*dzénte	-----
R47	*tsímpu	*dzínte	*λímbə
R52	-----	zínte	-----
R54	-----	-----	*yímbə
R56	-----	-----	*ímbə
R59	*tsimp	-----	----

The actual DR reflexes are timp, dinte and limba.

R27B Diphthongization of [ɛ]: [ɛ] > [ye] (P36A)

Formally:

$\emptyset > y / \text{ — } \overset{V}{[-\text{tense}]}$

Examples:

pétra > pyétra >> DR piatră 'stone'
 béne > byéne >> DR bine 'well'
 véne > vyéne >> DR vine '(s)he comes'
 téne > tyéne >> DR ține '(s)he holds'
 épa > yépa >> DR iapă 'mare'
 čélu > čyélu >> DR cer 'sky'
 péptu > pyéptu >> DR piept 'chest'
 déda > dyéda >> DR zadă 'larch tree'
 térra > tyérra >> DR țară 'earth'
 sélle > syélle >> DR șa 'saddle'

Let us now compare the following Late Latin forms with their modern Romance reflexes:

LL	tēne	vēne	vēntu	tēmpu	dēnte	bēne	kēlu
It	tiene	viene	vento	tempo	dente	bene	cielo
Ptg	tem	vem	vento	tempo	dente	bem	ceu
Sp	tiene	viene	viento	tiempo	diente	bien	cielo
Fr	tient	vient	vent	temps	dent	bien	ciel
Rom	ține	vine	vînt	timp	dinte	bine	cer

This chart illustrates that the diphthongization of [ɛ] to [ye] occurred in some environments in at least French, Spanish, Italian, and Romanian but apparently not in other Romances such as Portuguese. This raises the question of whether [ɛ] > [ye] applied during the Latin (or Balkan Romance) period and would therefore have applied before the individuation of the Romance dialects or afterwards and be therefore a parallel development in all those Romance dialects in which it occurred. It seems reasonable to assume that there was a tendency for CL E to open and lengthen [ɛ:] in the post-classical period, but that the actual creation of a diphthong did not occur until the various dialects were separated. This hypothesis is supported by the fact that the environments for this change are different in each Romance language. For a detailed discussion of diphthongization in the Romance languages see Otero:1985.

R28 Unstressed [o] > [u]

Formally,

$$\left[\begin{array}{c} \text{v} \\ +\text{back} \\ -\text{high} \\ -\text{low} \\ -\text{acc} \end{array} \right] > [+high]$$

Examples:

ómo > ómu >> DR om 'man; person'

kándo > kándu >> DR cînd 'when'

orrésko > urrésku >> DR urăsc 'I hate'

lyépore > lyépure >> DR iepure 'hare'

rrogáre > rrugáre >> DR a ruga 'to beg'

In R28, the Late Latin tendency for unstressed [o] to become [u] became more widespread (Sala 1976:212).

All Romanian dialects underwent this change. The southern dialects seem to maintain it with more consistency than does DR. In DR paradigms [o] was sometimes reestablished in unstressed position due to leveling. The present indicative forms of the verbs a putea 'to be able' and a dormi 'to sleep' illustrate this point.

Normal:	Leveled:
a putea	a dormi (instead of a durmi)
pot	dorm
poti	dormi
poate	doarme
putem	dormim (instead of durmim)
puteți	dormiți (instead of durmiti)
pot	dorm

The following rule, R29, deals with the initial stage of the appending of the postposed definite article, which is discussed in detail in Chapter 7.

R29A Loss of stress in the postposed forms of éllu

In what was to become the suffixed definite article only, the contemporary reflexes of CL ILLE lose their

stress. Formally,

e > [-acc] / # ____ in the postposed article

Examples:

lúpu éllu > lúpu ellu >> DR lupul [lúpu] 'the wolf'

kása élla > kása ella >> DR casa 'the house'

stélla élla > stélla ella >> DR steaua 'the star'

lúpi élli > lúpi elli >> DR lupii [lúpi] 'the wolves'

káse élle > káse elle >> DR casele 'the houses'

stélle élle > stélle elle >> DR stelele 'the stars'

lúpu ellúy > lúpu elluy >> DR lupului 'of/to the wolf'

káse ellyéy > káse ellyey >> DR casei 'of/to the house'

stélle ellyéy > stélle ellyey >> DR stelei 'of/to the
star'

lúpi ellóru > lúpi elloru >> DR lupilor 'of/to the
wolves'

káse ellóru > káse elloru >> DR caselor 'of/to the
houses'

stélle ellóru > stélle elloru >> DR stelelor 'of/to the
stars'

díe élla > díe ella >> DR ziua 'the day'

It is possible that the form elle served also as an
alternate modifier at this point for masculine singular
nouns ending in -e:

káne elle > kane elle >> DR cîinele 'the dog'

29B Apheresis and enclisis of the forms of [ellu]

In the suffixed definite article only, unstressed initial [e] > Ø and the resulting unstressed form becomes enclitic, i.e. it is appended to the preceding word.

Formally,

e > Ø / # ____ in the postposed article

Examples:

lúpu ellu > lúpullu >> DR lupul [lúpu] 'the wolf'

kása ella > kásalla >> DR casa 'the house'

stélla ella > stéllalla >> DR steaua 'the star'

lúpi elli > lúpilli >> DR lupii [lúpi] 'the wolves'

káse elle > káselle >> DR casele 'the houses'

stélle elle > stéllelle >> DR stelele 'the stars'

lúpu elluy > lúpulluy >> DR lupului 'to/of the wolf'

káse ellyey > kásellyey >> DR casei 'to/of the house'

stélle ellyey > stéllellyey >> DR stelei 'to/of the
star'

lúpi elloru > lúpilloru >> DR lupilor 'to/of the
wolves'

káse elloru > káselloru >> DR caselor 'to/of the
houses'

stélle elloru > stéllelloru >> stelelor 'to/of the
stars'

díe ella > dílla >> DR ziua 'the day'

káne elle > kánelle >> DR cîinele 'the dog'

R30 Preyodization of initial [e]: [e] > [ye]

Formally,

Ø > y / # ____ e in some words

Examples:

éste > yéste >> DR este [yéste] 'is'

éllu > yéllu >> DR el [yel] 'he'

élla > yélla >> DR ea [ya] 'she'

élli > yélli >> DR ei [yey] 'they m'

élle > yélle >> DR ele [yéle] 'they f'

éu > yew >> DR eu [yew] 'I'

esíre > yesíre >> DR a ieși 'to go out'

e > ye >> DR e [ye] 'is'

éra > yéra >> DR era [yerá] '(s)he was imperf'

and in the remaining forms of the imperfect of a fi 'to be'.

This preyodization is probably due to Slavic influence. It is interesting to note that modern Romanian and Serbo-Croatian both have two forms that translate 'is' and that the Romanian forms are identical to the Serbo-Croatian forms.

DR este, e [yéste], [ye]

SC jeste, je [yéste], [ye]

In Bulgarian, however, the Slavic language with which Romanian has had the most contact during its development, the one word for 'is' is non-preyodized [e].

kópsa > kópsa >> DR coapsă 'thigh'
 mólla > mól̥lla >> DR moală 'soft f'
 yélla > yé̥lla >> DR ea [ya] 'she'
 yélle > yé̥lle >> DR ele [yéle] 'they f'
 stélla > sté̥lla >> DR stea [steá] 'star'
 stélle > sté̥lle >> DR stele 'stars'
 pénna > pé̥nna >> DR pană 'feather'

but

byene remains byene >> DR bine 'well'
 bona remains bona >> DR bună 'good'

R27B had to apply before R31A. If R31B had preceded R27B, it would not have been possible to explain the fact that, in Romanian, the [ɛ] that results from metaphony evolves to [ea], whereas the [ɛ] resulting from Latin E, AE, becomes [ye].

R31B Diphthongization of lax vowels: [ɛ]/[ɔ] > [ea]/[ɔa]

Formally,

$\emptyset > a / \begin{matrix} V \\ [-tense] \end{matrix} \quad \text{---}$

then

$\begin{matrix} V \\ [-tense] \end{matrix} > [-syl]$

Examples:

négra > neágra >> DR neagră 'black f'
 négre > neágre >> DR negre 'black f pl'

s^éra > seára >> DR seară 'evening'
 m^ésa > meása >> DR masă 'table'
 p^éнна > peánna >> DR pană 'feather'
 nev^ésta > neveásta >> DR nevastă 'wife'
 kam^ésya > kameásya >> DR cămaşă 'shirt'
 st^éllalla > steállalla >> DR steaua 'the star'
 dy^éda > dyeáda >> DR zadă 'larch tree'
 py^étra > pyeátra >> DR piatră 'stone'
 ty^érra > tyeárra >> DR ţară 'earth'
 sy^élla > syeálla >> DR şa 'saddle'
 sy^épte > syeápte >> DR şapte 'seven'
 dy^éce > dyeáce >> DR zece 'ten'
 y^éste > yeáste >> DR este [yéste] 'is'
 y^élla > yeálla >> DR ea [ya] 'she'
 y^élle > yeállle >> DR ele [yéle] 'they f pl'
 m^ólle > moállle >> DR moale 'soft f pl'
 k^óda > koáda >> DR coadă 'tail'
 d^ómna > doámna >> DR doamnă 'lady'

Examples in the southern dialects:

CL FOETA >> AR feátə 'girl'
 CL DIGITU- >> AR deádzit >> MR zeázit 'finger'
 CL NIGRA >> AR neágrə 'black f'
 CL SITE- >> MR seáti 'thirst'
 CL ROTA >> AR, MR roátə 'wheel'
 CL SOLE >> AR soáre >> MR soári 'sun'

IR has [ɛ] and [ɔ] where all the other dialects have [ea] and [oa], which seems to indicate that R31B did not apply in IR. According to Puscariu such is not the case: R31B did apply in IR but a subsequent rule changed [ea] and [oa] back to [ɛ] and [ɔ] (Vasiliu 1968:66).

CL NIGRA >> IR nɛ́grə 'black f pl'

CL FOETAE >> IR féte 'girls'

CL CODA >> IR kóde 'tail'

CL NOKTE >> IR nópte 'night'

Slavic words also underwent R31:

OS kósa >> DR coasă 'scythe'

As in R27B, the diphthongization of stressed lax mid-vowels in R31B may be due to a lengthening and differentiation process as illustrated in the following derivation:

Latin	nígra	dóm(i)na
R13F	negra	-----
R31A [-tense]	nɛ́gra	də́mna
[+long]	nɛ́́gra	dɔ́́mna
Diphthong	neégra	doó́mna
[+low]	neágra	doámna
DR	neagra	doamna

R32 Displacement of accent and loss of syllabicity:

[éa] > [eá]

Formally,

$\begin{bmatrix} e \\ +acc \end{bmatrix}$	a	>	[-acc]	[+acc]
1	2		1	2

which concomitantly shifts the stress to the low vowel.

Examples:

méa > meá >> DR mea 'my'

avéa > aveá >> DR avea '(s)he had imperf'

[éa] became [eá] with the transformation of [e] into a semi-vowel and the shifting of the accent onto [a] (Vasiliu 1968:33).

R33 Raising of [a] and [o] before a non-geminate nasal

Formally,

$$\left\{ \begin{array}{l} a > \text{ə} \\ o > u \end{array} \right\} / \text{---} \begin{bmatrix} C \\ +nas \end{bmatrix} V$$

Examples:

káнду > kəнду >> DR cînd 'when'

betrán > betrən >> DR bătrîn 'old man'

bónu > búnú >> DR bun 'good m'

mónte >> mún-te = DR munte 'mountain'

kómparu > kúmparu >> DR cumpăr 'I buy'

but

ánnu remains ánnu

đoámna remains đoámna

All dialects underwent R33:

LL bonu- >> DR, AR, MR bun; IR bur

LL monte- >> DR, AR munte; MR munti

LL komparo >> DR, MR kúmpær, AR akúmpær; MR kúmparu

Later, in DR, R47B raises [e] to [i] in this environment.

The same rule also applies in southern AR.

R34 Rhotacism: non-geminate [l] > [r] between vowels

Formally,

l > r / V ____ V

Examples:

soále > soáre = DR soare 'sun'

mélu > méru >> DR măr 'apple'

pélu > péru >> DR pǎr 'hair'

myéle > myére = DR miere 'honey'

kále > káre = DR care 'which'

sále > sáre = DR sare 'salt'

This rule applied to words of Latin origin in all dialects:

CL solem >> DR soáre; MR soári; IR sóre

BR melum >> DR măr; MR, IR mer

It did not apply to words of Slavic origin:

OS kolaci >> DR, AR colac 'twist'

OS zali >> DR, AR jale 'grief'

OS tocilo >> DR tocilă 'grindstone'

which suggests two possibilities: 1) the rule had already applied and was no longer in effect when OS words penetrated into CR, or 2) the intervocalic OS [l] was somehow phonetically different from the BL [l], perhaps more akin to the BL [ll] ([+ velar]), which did not undergo rhotacism.

R35A [ll] > [w] before unstressed [a]

Formally,

ll > w / ^a
 [-acc]

Examples:

yeálla > yeáwa >> DR ea [yá] 'she'

steálla > steáwa >> DR stea [steá] 'star'

steállalla > steáwawa >> DR steaua 'the star'

kásalla > kásawa >> DR casa 'the house'

vúlpella > vúlpewa >> DR vulpea 'the fox'

medúlla > medúwa >> DR măduvǎ 'marrow'

pílla > píwa >> DR piuă 'fulling mill'

Note that this rule does not affect words like CR medulláre in which the [a] is stressed. CR medulláre >> DR mădular 'limb'. [ll] probably passed through a velarized state before it became [w]. Velar l's have become vocalized in

many languages, among them Polish, Serbo-Croatian, Portuguese, and the Cockney variety of English.

R35B Degemination of [ll]

Formally,

ll > l

Examples:

kaállu > kaálu >> DR cal 'horse'

kaállullu > kaálulu >> DR calul [kálu] 'the horse'

moállle > moále = DR moale 'soft f pl'

steálllelle > steálele >> DR stelele 'the stars'

gallína > galína >> DR găină 'hen'

lúpilli > lúpili >> DR lupii [lúpi] 'the wolves'

mólli > móli >> DR moi 'soft m pl'

R36A Loss of secondary wau: [Vwa] > [Va] when neither vowel is stressed

Formally,

$$w > \emptyset / \begin{matrix} V \\ [-acc] \end{matrix} \quad \text{---} \quad \begin{bmatrix} V \\ +low \\ -acc \end{bmatrix}$$

Examples:

kásawa > kásaa >> DR casa 'the house'

steáwawa > steáwaa >> DR steaua 'the star'

vúlpewa > vúlpea >> DR vulpea 'the fox'

R36B F sg n/a def art [a] > [+long] in all f sg nouns

Formally,

[a] > [+long] / ____ # in all articulated f sg nouns in
the nominative/accusative case

Examples:

zíwa > zíwaa >> DR ziua 'the day'

vúlpea > vúlpeaa >> DR vulpea 'the fox'

This is in reality a morpho-phonologic rule, since the
conditions are morphologic.

R36C [úwa] > [úva]

Formally,

w > v / u ____ a

Example:

medúwa > medúva >> DR măduvă 'marrow'

R37A Raising of non-initial, non-geminate, unstressed [a]

Formally,

$$\left[\begin{array}{c} \text{v} \\ +\text{low} \\ -\text{acc} \\ -\text{long} \end{array} \right] > \quad / \quad \text{c} \quad \underline{\quad}$$

Examples:

kása > kásə = DR, AR casă 'house'

kameásya > kameásyə >> DR cămaşă 'shirt'

parénte > pərénte >> DR părinte 'parent'

sanetáte > sənetáte >> DR sănătate 'health'

úna > únə = MR únə 'one'

kumparáre >> MR kumperó 'to buy'

This rule applied to words of Slavic origin as well:

platíti > plətíti >> DR a plati 'to pay'

R37B Degemination of geminate vowels

Formally,

$V_i V_i > V_i$

Examples:

kásaa > kása >> DR casa 'the house'

dzyíwaa > dzyíwa >> DR ziua 'the day'

purtáa >> purtá = DR purta '(s)he carried imperf'

kaálu > kálu >> DR cal 'horse'

beere > bere >> DR a bea 'to drink'

According to Vasiliu, [aa] contracted to [a] (1968:31).

With this rule the feminine singular definite article takes the phonetic shape it presently has in DR as does the third person singular of the imperfect for the first conjugation.

R38 [eá] > [eá]

[e] becomes a glide before [a]

$e > [-\text{syl}] / ___ a$

Examples:

vúlpea > vúlpea = DR vulpea 'the fox'

The hypothesis that the sequence [ea] became a diphthong is supported by its presence as such in each

Romanian dialect today (Vasiliu 1968:157).

R39 Lowering of unstressed internal [i]

Formally,

i > e / C ____ C
[-acc]

Examples:

citáte > cetáte = DR cetate 'fortress'

tičóne > tečóne >> DR tăciune 'ember'

diréĵe > deréĵe >> DR a drege 'to mend'

vićínu > većínu >> DR vecin 'neighbor'

R39 is necessary at this point in order to bleed R40.

R40 Diphthongization of [i]: [i] > [yi]

Formally,

∅ > y / ____ i

Examples:

díku > dyíku >> DR zic 'I say'

línu > lyínu >> DR in 'flax'

vínu > vyínu >> DR vin 'wine'

mutíre > mutyíre >> DR a amuĵi 'to silence'

fáčeti > fáčetyi >> DR faceĵi 'you do pl'

R34 was proposed by Puşcariu (1927) and reintroduced by Sala (1976). The hypothesis is that the yod created by the diphthongization of [i] provided the same condition for palatalization and affrication of alveolars as did the yod

kəməásyə > kəməášyə >> DR cămaşă 'shirt'

R41B Absorption of yod after a palatal segment

Formally,

y > Ø / $\begin{matrix} \text{C} \\ +\text{high} \end{matrix}$ —

Examples:

lyépure > lépure >> DR iepure 'hare'

pályə > pálə >> DR pai(e) 'straw'

lyínu > línu >> DR in 'flax'

kəlkáñyu > kəlkáñu >> DR călcâi 'heel'

kəpetáñyu > kəpetáñu >> DR căpătîi 'head of the bed'

kúñyu > kúñu >> DR cui 'nail'

vyíñyə > vyíñə >> DR vie 'vineyard'

šyésu > šésu >> DR şes 'plain'

šyęápte > šęápte >> DR şapte 'seven'

šyí > ší >> DR şi 'and'

yešyíre > yešíre >> DR a ieşi 'to to out'

kəməášyə > kəməášə >> DR cămaşă 'shirt'

R42A Affrication of alveolar stops before yod:

[t] > [ts], [d] > [dz]

Formally,

$\begin{bmatrix} -\text{cont} \\ +\text{ant} \\ +\text{cor} \\ -\text{del rel} \end{bmatrix}$ > [+del rel] / — y

Examples:

tyésu > tsyésu >> DR ȣes 'I weave'
 tyéne > tsyéne >> DR ȣine '(s)he holds'
 tyéstu > tsyéstu >> DR ȣest 'griddle'
 pətyíre > pətsyíre >> DR a păȣi 'to experience'
 kutyítu > kutsyítu >> DR cuȣit 'knife'
 dyeáče > dzyeáče >> DR zece 'ten'
 dyíku > dzyíku >> DR zic 'I say'
 awdyíre > awdzyíre >> a auzi 'to hear'

Note that [ts] and [dz] had already developed during the BR period with the application of R18-R19 before R27B (ȣ > ye) and R40 (i > yi), both of which had to apply to produce the yod which makes the affrication in R42A possible.

R42B Absorption of yod after an affricate

Formally,

y > ø / [+del rel] ____

Examples:

tsyésu > tsésu >> DR ȣes 'I weave'
 tsyéne > tséne >> DR ȣine '(s)he holds'
 tsyéstu > tséstu >> DR ȣest 'griddle'
 pətsyíre > pətsíre >> DR a păȣi 'to experience'
 kutsyítu > kutsítu >> DR cuȣit 'knife'
 glátsyə > glátsə >> DR gheaȣă 'ice'
 dzyeáče > dzeáče >> DR zece 'ten'

dzýíku > dzíku >> DR zic 'I say'

audzyíre > audzíre >> DR a auzi 'to hear'

R43 Palatalization of [l] after a velar consonant

Formally,

$$l > \begin{bmatrix} +\text{high} \\ -\text{ant} \end{bmatrix} / \begin{bmatrix} -\text{cont} \\ -\text{ant} \\ -\text{cor} \end{bmatrix} \text{ —}$$

Examples:

óklu > ókλu >> DR ochi 'eye'

véklu > vékλu >> DR vechi 'old m'

véklə > vékλə >> DR veche 'old f'

glátsə > gλátsə >> gheață 'ice'

With the application of R43, the partial assimilation of [l] to the point of articulation of the preceding velar occurs.

R44 Fronting of [ə] after [λ] and [ñ]

Formally,

$$ə > e / \begin{bmatrix} -\text{cont} \\ +\text{high} \\ -\text{str} \end{bmatrix} \text{ —}$$

Examples:

vékλə > vékλe >> DR veche 'old f'

páλə > páλe >> DR paie 'straw'

vyíñə > vyíñe >> DR vie 'vineyard'

R45A Assimilation: [s] > [š] before [č]

Formally,

$$s > \begin{bmatrix} -\text{ant} \end{bmatrix} / \text{---} \begin{bmatrix} +\text{del rel} \\ -\text{ant} \\ +\text{cor} \end{bmatrix}$$

Examples:

sčiw > ščiw >> DR știu 'I know'

kresči > krešči >> DR crești 'you grow'

R45B Dissimilation: [č] > [t] after [š]

Formally,

$$\begin{bmatrix} +\text{del rel} \\ -\text{ant} \\ +\text{cor} \end{bmatrix} > \begin{bmatrix} -\text{del rel} \\ +\text{ant} \end{bmatrix} / \text{š} \text{---}$$

Examples:

ščiw > štíw >> DR știu 'I know'

krešči > krešti >> DR crești 'you grow sg'

5.3 The Phonetic Inventory of Common Romanian

The sounds [z], [ž] and [h] entered the phonetic inventory of Romanian during the CR period via borrowings from OS (Sala 1976:249).

After the application of R26 - R45, the phonetic inventory of CR included the following:

Vowels: [a], [e], [i], [o], [u], [ə]

Glides: [ɛ], [y], [ɔ], [w], [h]

Consonants: [p], [b], [m], [f], [v], [t], [d], [n], [nn],
[l], [r], [rr], [ts], [dz], [s], [z], [č], [ž],

[ñ], [λ], [š], [ž], [k], [g], [ŋ].

CHAPTER 6

THE DEVELOPMENT OF DACO-ROMANIAN (11TH CENTURY TO THE PRESENT)

6.1 The Term 'Daco-Romanian'

According to Vasiliu (1968:123), there was never a common Daco-Romanian dialect from which Muntenian, Moldavian and others developed. Daco-Romanian simply designates the area in which these dialects developed independently from CR. Due to their close proximity, however, there is little doubt that they have had an adstrat effect on one another.

Nonetheless, the rules that derive standard Romanian from the Muntenian dialect are to a great extent shared by all the Daco-Romanian dialects. The difference seems to be the order in which they applied. The following are rules and the order by means of which standard Romanian (DR) can be derived from CR. Parenthetical references such as (V10) refer to Vasiliu's 'schimbări fonetice' (sound changes) in Fonologia istorică a dialectelor dacoromâne.

6.2 Rules

R46A [e],[ə] > [ɛ],[ǣ] after a labial and before a syllable containing a back vowel (V10)

Formally,

$$\begin{bmatrix} \text{-cons} \\ \text{-back} \\ \text{-high} \end{bmatrix} > [+back] / \begin{bmatrix} \text{C} \\ \text{+ant} \\ \text{-cor} \end{bmatrix} \text{ — Co } \begin{matrix} \text{V} \\ [+back] \end{matrix}$$

Examples:

méru > mǣru >> DR mǣr 'apple'
 péru > pǣru >> DR pǣr 'pear tree'
 védu > vǣdu >> DR vǣd 'I see'
 vénə > vǣnə >> DR vǣnǣ 'vein'
 véntu > vǣntu >> DR vǣnt 'wind'
 fénu > fǣnu >> DR fǣn 'hay'
 bešíkə > bǣšíkə >> DR bǣšicǣ 'bladder'
 vertúte > vǣrtúte >> DR vǣrtute
 kǣmǣašə > kǣmǣášə >> DR cǣmašǣ 'shirt'
 feátə > fǣátə >> DR fatǣ 'girl'
 mǣásə > mǣásə >> DR masǣ 'table'
 neveástə > nevǣástə >> DR nevastǣ 'wife'
 peánnə > pǣánnə >> DR panǣ 'feather'

but

méri remains méri >> DR meri 'apples'
 péri remains péri >> DR peri 'pear trees'
 vézi remains vézi >> DR vezi 'you see sg'
 veáde remains vǣáde >> DR vede '(s)he sees'

većínu remains većínu >> DR vecin 'neighbor'
 meáse remains meáse >> DR mese 'tables'
 peáinne remains peáinne >> DR pene 'feathers'
 beáire remains beáire >> DR bere 'drinking'

R46B Loss of [ə] before [a]: [əa] > [a]

Formally,

ə > Ø / ____ a

Examples:

kəməášə > kəmásə >> DR cămașă 'shirt'
 fəátə > fátə >> DR fată 'girl'
 məásə > másə >> DR masă 'table'
 nevəástə > nevástə >> DR nevastă 'wife'
 peáanna > pannə >> DR pană 'feather'

R47A Raising of [ə] before [n] or [m]CV (V59)

Formally,

ə > [+high] / ____ {ⁿ_{mCV}}

Examples:

vəná > vînă = DR vînă 'vein'
 vəntu > vîntu >> DR vînt 'wind'
 kəntu > kîntu >> DR cînt 'I sing'
 kəmpu > kîmpu >> DR cîmp 'field'
 əntrégu > întregu >> DR întreg 'whole'

R47B Raising of [e] before [n] (V63)

Formally,

$$e > [+high] / \text{ — } \left[\begin{array}{c} C \\ +nas \end{array} \right]$$

Examples:

byéne > byíne >> DR bine 'well'

lémbə > límbə >> DR limbă 'tongue; language'

témpu > tímpu >> DR timp 'time'

It is possible that some words were inherited from Late Latin with [e] already raised to [i] in this environment.

R48 Degemination of [nn]: [nn] > [n]

Formally,

$$nn > n$$

Examples:

ánnu > ánu >> DR an 'year'

pánnə > pánə = DR pană 'feather'

peáanne > peáne >> DR pene 'feathers'

R49 Raising of [ə] before [r,s] plus consonant

(Vasiliu 1968:63)

Formally,

$$ə > ɪ / \text{ — } \left\{ \begin{array}{c} r \\ s \end{array} \right\}^C$$

Examples:

vərtúte > vɛrtúte = DR vîrtute 'virtue'

gə́skə > gɛ́skə = DR giscă 'goose'

kəstigáre > kĭstigáre >> DR a cĭştiga 'to win'

R50 Backing of front vowels after [rr] (V13)

Formally,

V > +back / rr ____

Examples:

rrēw > rrəw >> DR rău 'bad'

urrésku > urrésku >> DR urăsc 'I hate'

rrīpə > rrĭpə >> DR rĭpa 'cliff, gully'

rrīmu > rrĭmu >> DR rĭm 'I rout'

rrīw > rrĭw >> DR rĭu 'river'

rrĭde > rrĭde >> DR rĭde '(s)he laughs'

R51 Simplification of [rr] to [r]

Formally,

rr > [-rep]

Examples:

rrəw > rəw >> DR rău 'bad'

urrésku > urésku >> DR urăsc 'I hate'

urrĭtu > urĭtu >> DR urĭt 'ugly'

rrĭde > rĭde >> DR rĭde '(s)he laughs'

R52A Deaffrication of voiced alveolar affricate:

[dz] > [z]

Formally,

$$\begin{bmatrix} +\text{del rel} \\ +\text{ant} \\ +\text{voice} \end{bmatrix} > [-\text{del rel}]$$

Examples:

dzyi > zyi >> DR zi 'day'

védzyi > vézyi >> DR vezi 'you see'

prîndzu > prînz >> DR prînz 'lunch'

R52B Deaffrication of voiced palatal affricate:

[ʝ] > [ʒ] before [o] or [u] (V47)

Formally,

$$\begin{bmatrix} -\text{son} \\ +\text{del rel} \\ -\text{ant} \end{bmatrix} > [+cont] / \text{---} \begin{bmatrix} \text{V} \\ +\text{round} \end{bmatrix}$$

Examples:

ǵóku > ʒóku >> DR joc 'game'

aǵutáre > aʒutáre >> DR a ajuta 'to help'

ǵukáre > ʒukáre >> DR a juca 'to play'

[z] and [ʒ] had already entered the language during the CR period via Slavic borrowings (Sala 1976:249).

R57A and R57B applied only in the Daco-Romanian dialects. The southern dialects have retained [dz] and [ʝ].

R53A Monophthongization of [ea] after [š] or [ts]

Formally,

$\bar{e}a > a / \left\{ \begin{smallmatrix} \bar{s} \\ ts \end{smallmatrix} \right\} \text{ — (sporadically)}$

Examples:

š $\bar{e}$ ápte > šápte >> DR šapte 'seven'

š $\bar{e}$ árpe > šárpe >> DR šarpe 'snake'

tš $\bar{e}$ árə > tsárə >> DR țară 'land'

This rule is necessary at this point to bleed R53B. Otherwise the forms for 'seven', 'snake', etc., would be *š $\bar{e}$ pte, *š $\bar{e}$ rpe in DR.

R53B Monophthongization of [ea] before a syllable containing [e] (V65)

Formally,

$\bar{e}a > e / \text{ — } \$ \text{ Co } e$

Examples:

m $\bar{e}$ áse > mése = DR mese 'tables'

l $\bar{e}$ áje > léje = DR lege 'law'

v $\bar{e}$ áde > véde = DR vede '(s)he sees'

st $\bar{e}$ ále > stéle = DR stele 'stars'

p $\bar{e}$ áne > péne = DR pene 'feathers'

z $\bar{e}$ áče > zéče = DR zece 'ten'

b $\bar{e}$ áre > bére = DR bere 'drinking'

In sixteenth century texts, in words in which [ea] preceded [e], the former vowel was written either $\bar{e}$, $\bar{e}$, or $\bar{e}$.

Ț represented the simple vowel [e]; Ț and Ț̣ represented the diphthong [ɛa]. Further complicating the issue is the fact that Ț and Ț̣ appear in words in which [ɛa] never existed. Vasiliu believes that this graphic alternation indicates that [e] and [ɛa] were in free variation in this environment, and that [e] eventually supplanted [ɛa] here sometime after the sixteenth century (Vasiliu 1968:154).

The final syllable -re of Romanian infinitives must have been deleted prior to the application of R56. Otherwise, the diphthong [ɛa] in verbs like 'a bea', 'a putea', etc., would have monophthongized giving *bere > *a be.

R54 Delateralization of [λ] (V61, V64)

Formally,

λ > y

Examples:

lépure > yépure >> DR iepure 'hare'

gə́línə > gə́yínə >> DR găină 'hen'

vékλu > vékyu >> DR vechi 'old m'

vékλe > vékye = DR veche 'old f'

gλátsə > gyátsə = DR gheață 'ice'

káλiλi > káyiyi > DR caii 'the horses'

But AR and MR conserve the [λ] (Vasiliu 1968:115):

CL GALLINA >> AR, MR [gə́línə]

R55 Denasalization of [ñ] (V60)

Formally,

$$\begin{bmatrix} +\text{son} \\ -\text{ant} \\ +\text{cor} \end{bmatrix} > \begin{bmatrix} -\text{nas} \\ -\text{cons} \\ -\text{lat} \end{bmatrix}$$

Examples:

spúñi > spúyi >> DR spui 'you say'

púñi > púyi >> DR pui 'you put'

kúñu > kúyu >> DR cui 'nail'

DR forms such as buni, bani, cîni, etc., are back-formations based on the singular.

R56 Loss of yod before [i]

Formally,

$$y > \emptyset / __ i$$

Examples:

gəyínə > gəínə = DR găină 'hen'

spúyi > spúy = DR spui 'you say'

byíne > bíne = DR bine 'well'

vyínu > vínu > DR vin 'wine'

fácetsyi > fáčetsi >> DR faceți 'you do pl'

káiyi > cáii >> DR cai 'the horses'

vyíye > víye = DR vie 'vineyard'

R57 [yea] > [ya]

Formally,

$$e > \emptyset / y __ a$$

Examples:

pyɛátrə > pyátrə >> DR piatră 'stone'

yɛárbə > yárbə >> DR iarbă 'grass'

yɛárnə > yárnə >> DR iarnă 'winter'

R58 Assimilation: word-final [ə] > [e] after [i]

Formally,

ə > -back / i ____ #

Examples:

pələríə > pələríe = DR pălărie 'hat'

kutíə > kutíe = DR cutie 'box'

R59A Devoicing of word-final [u] and [i] except after a
consonant plus a liquid (V27a,b)

Formally,

V > [-voice] / X ____ #
[+high]

where X is not the sequence consonant + liquid

Examples:

lúpu > lupɥ >> DR lup 'wolf'

brádu > bradɥ >> DR brad 'fir'

vékýu > vekýɥ >> DR vechi 'old m'

lúpi > lúpy = DR lupi 'wolves'

fáčetsi > fáčetsɥ = DR faceți 'you do pl'

but

lúkru remains DR lucru 'thing'

péntru remains DR pentru 'for'

ámplu remains DR amplu 'ample'

áflu remains DR aflu 'I find'

R59B Loss of word-final voiceless [w]

Formally,

w > ∅

Examples:

kúyw > kúy = DR cui 'nail'

vekyw > veki = DR vechi 'old m'

kálw > kál = DR cal 'horse'

kálulw > kálul >> DR calul [kálu] 'the horse'

This rule also applied to some degree in MR and IR but Vasiliu correctly sees it applying after the split. It seems to be a case of tendential change before separation becoming widespread, independently, after separation.

R59C Word-final [ii] > [i]

Formally,

ii > i / ____ #

Examples:

lúpii > lúpi = DR lupii [lúpi] 'the wolves'

R60 Loss of final [l] of m sg definite article

Formally,

l > ∅ / ____ # in m sg definite article

Examples:

kálul > DR calul [kálu] 'the horse'

In all dialects this is probably due to reanalysis of forms such as calului to calu + lui, since lui had become an independent preposition meaning 'of' (lui Radu 'of Radu').

CHAPTER 7

THE ENCLITIC DEFINITE ARTICLE

7.1 Introduction

I have chosen to elaborate on the phonological evolution of the definite article, which in Romanian is postposed, mainly because I have not been at all satisfied with the sketchy, confusing and often contradictory explanations of this intriguing phenomenon provided in handbooks and scholarly articles.

Enclisis of the definite article is one of the features which sets Romanian apart from the other Romance languages. Other Indo-European languages which have a postposed definite article are Bulgarian, Albanian and the Scandinavian languages. In the 19th century, attempts were made to trace this development in Romanian to contact with Scandinavian, but there was no concrete evidence to support such a theory, and it was soon entirely abandoned.

It is generally accepted, however, that Romanian, Albanian and Bulgarian each developed an enclitic definite article due to some sort of mutual influence. To begin with, these three languages, along with Serbo-Croatian and Greek are part of a linguistic community in which genetically distant languages have become typologically very

similar, identical in many aspects, through historical co-territoriality and bi- (possibly tri-) lingualism.

It is possible that both Albanian and Romanian enclisis was due to a substratum influence common to both, perhaps Thracian and/or Illyrian. Another theory has it that Latinized Albanians were among the ancestors of the Romanians, and that an Albanian substratum therefore caused the enclisis in Romanian. Some linguists have theorized that enclisis was a purely internal development in both Albanian and Romanian. In any event, it seems certain that enclisis in Bulgarian, a much later development, was mainly due to the adstrat influence of Romanian. Neither of the other two languages of the Balkan linguistic community, Serbo-Croatian and Greek, has a postposed definite article.

7.2 The Form of the Article in Modern Standard Romanian

Here are the basic articulated forms in DR with the corresponding non-articulated forms for comparison:

Masculine:

<u>un lup</u>	[un lúp]	'a wolf'
<u>lupul</u>	[lúpu]	'the wolf'
<u>lupi</u>	[lúpy]	'wolves'
<u>lupii</u>	[lupi]	'the wolves'
<u>unui lup</u>	[unuy lúp]	'to/of a wolf'
<u>lupului</u>	[lúpuluy]	'to/of the wolf'

<u>unor lupi</u>	[unor lúpy]	'to/of some wolves'
<u>lupilor</u>	[lúpilor]	'to/of the wolves'

For singular masculine nouns ending in e, the n/a article is le (instead of ul). All other endings are identical to those for lup.

<u>un cîine</u>	[un káyne]	'a dog'
<u>cîinele</u>	[káynele]	'the dog'

It is not certain when or why this special form developed. Most linguists agree, however, that the survival of the CL form ILLE here is highly improbable.

Feminine:

<u>o casă</u>	[o kása]	'a house'
<u>casa</u>	[kása]	'the house'
<u>case</u>	[káse]	'houses'
<u>casele</u>	[kásele]	'the houses'
<u>unei case</u>	[uney káse]	'to/of a house'
<u>casei</u>	[kásey]	'to/of the house'
<u>unor case</u>	[unor káse]	'to/of some houses'
<u>caselor</u>	[káselor]	'to/of the houses'

Another important feminine paradigm developed from CL words ending in ella (margella, sella, stella, vitella, etc.):

<u>o stea</u>	[o steá]	'a star'
<u>steaua</u>	[steáwa]	'the star'
<u>stele</u>	[stéle]	'stars'
<u>stelele</u>	[stélele]	'the stars'

<u>unei stele</u>	[uney stéle]	'to/of a star'
<u>stelei</u>	[stéley]	'to/of the star'
<u>unor stele</u>	[unor stéle]	'to/of some stars'
<u>stelelor</u>	[stélelor]	'to/of the stars'

7.3 The Traditional Hypothesis

It is generally accepted that the definite article in Romanian stems from post-classical Latin forms íllu, ílla, ílli, ílle for the nominative/accusative, and illúi, illéi, illóru for the genitive/dative.¹³ Many if not most handbooks and linguists also accept the hypothesis that the article developed proclitically and was subsequently appended to nouns and adjectives as illustrated in these incomplete, unattested, hypothetical derivations:

Masculine, singular, n/a:

illu lupu > alu lupu > lu lupu > lupu-lu > lupul >
DR lupul [lúpu] 'the wolf'

Masculine, singular, g/d:

illuy lupu > aluy lupu > luy lupu > lupu-luy >
DR lupului [lúpuluy] 'to/of the wolf'

Masculine, plural, n/a:

illi lupi > ali lupi > li lupi > lupi-li > lupili >
DR lupii [lúpi] 'the wolves'

Masculine, plural, g/d:

illoru lupi > aloru lupi > loru lupi > lupi-loru >
DR lupilor [lúpilor] 'to/of the wolves'

Feminine, singular, n/a:

illa kasa > awa kasa > a kasa > kasə-a > kasa >

DR casa [kasa] 'the house',¹⁴

Feminine, singular, g/d:

illɛy kasa > aɛy kasa > ɛy kasa > kasa-ɛy >

kasə-ey > kasei > DR casei [kásei] 'to/of the house'

Feminine, plural, n/a:

ille kase > ale kase > le kase > kase-le > kasele

DR casele [kásele] 'the houses'

Feminine, plural, g/d:

illoru kase > aloru kase > loru kase > kase-loru >

kaselor > DR caselor [káselor] 'of/to the houses'

Feminine, singular, n/a ending in -ella:

stella > stewa > stea > steá > DR stea [steá] 'star'

illa stella > awa stewa > a stea > a steá > steá-a >

DR steaua [steáwa] 'the star'

ille stelle > ale stele > le stele > stele-le > DR

stelele 'the stars'

According to Phillippide (1927:407 ff.) the article is found in both proclitic and enclitic position. In some dialects proclesis and enclisis seem to be in free variation in some cases although it is not at all certain what this free variation indicates with respect to the development of the definite article.

The chrestomathies, which begin with the first known DR

text (1521), record only enclitic articles. There are, however, instances in them of the proclitic use of the particles lu and lui. Although undeniably the reflexes of forms of Latin ILLE, these particles may alternatively be perceived and classified as prepositions as the unbound particle lui functions in contemporary DR: casa lui Radu 'the house of Radu, Radu's house'.

There are linguists who favor enclisis: Kr. Sandfeld (1930:165-173) doubts that the article was originally proclitic in Romanian, and Povl Skårup (1970:70) hypothesizes that the article began as enclitic in all the Romance languages and that only Romanian maintained it as enclitic.

7.4 An Organic Hypothesis

Upon inspecting the entire gamut of phonological rules, it became apparent to me that, with the exception of the feminine singular nominative/accusative forms, all the articles were derivable in enclitic position via the general historical rules of the language with the addition of only one article-specific rule and an instance of haplology. This led to a desire to explore the possibility that the feminine, singular nominative/accusative forms had also developed enclitically and in accordance with the established phonological rules beginning in the Common

Romanian period (i.e. organically).

To my knowledge, no one had ever attempted to trace the development of the present-day Romanian forms from their Latin etyma in a rigorous, rule-by-rule derivation, perhaps since no one had ever seriously tried to state and sequence historical phonological rules for that purpose.

As shown in the derivations below, all masculine forms and all feminine forms other than the feminine singular nominative/accusative forms are easily derived with the rules established for the CR and DR periods with the addition of one article-specific rule (R29 loss of accent, apheresis and enclisis) and one instance of haplology, neither of which affects other words in the language.

Masculine forms:

BR	lúpu éllu	lúpu ellúi	lúpi élli	lúpi ellóru
R29A	lúpu ellu	lúpu elluy	lúpi elli	lúpi ellóru
R29B	lúpullu	lúpulluy	lúpilli	lúpilloru
R35B	lúpulu	lúpuluy	lúpili	lúpiloru
R40	-----	-----	lúpyilyi	lúpyiloru
R41A	-----	-----	lúpyiλyi	-----
R41B	-----	-----	lúpyiλi	-----
R54	-----	-----	lúpyiyi	lúpiloru
R56	-----	-----	lúpii	-----
R59	lúpul	-----	-----	lúpilor
R59C	-----	-----	lúpi	-----
R60	lúpu	-----	-----	-----
DR	'lupul'	'lupului'	'lupii'	'lupilor'

Plural and oblique feminines, Type I:

BR	káse élle	káse elléy	káse ellóru
R27B	-----	káse ellyéy	-----
R29A	káse elle	káse ellyey	káse elloru
R29B	káselle	kásellyey	káselloru
R35B	kásele	káselyey	káseloru
R41A	-----	káseleyey	-----
R41B	-----	káseley	-----
R54	-----	káseyey	-----
R59	-----	-----	káselor
DR	casele	casei	caselor

'the houses' 'of the house' 'of the houses'

Philippide records forms similar to the singular genitive/dative form káseley in AR (doámna cásil'ei) among many which seem to support the steps of these derivations (1927:408).

I propose that the DR form [káseyey] became [kásey] via haplology, which is supported by the fact that in DR ei [yey] is also an independent possessive adjective meaning 'her' which could have helped bring about the haplology:

DR casei 'of the house' (not *káseyey)

DR casei ei 'of her house' (not *káseyey yey)

Besides, one [ey] is sufficient to carry the morphological data.

Plural and oblique feminines, Type II:

Another basic form of the feminine enclitic definite article developed from the evolution of feminine nouns ending in -ella, such as stella, margella, vitella, etc., and is now the form for any feminine noun ending in a stressed final vowel. The articulated plural and oblique forms of this second type of feminine developed exactly like Type I above. The paradigm is given here to facilitate comparison with forms below.

BR	stéлле elléy	stéлле éлле	stéлле ellóru
R27B	stéлле ellyéy	-----	-----
R29A	stéлле ellyey	stéлле elle	stéлле elloru
R29B	stéллеllyey	stéллеlle	stéллеlloru
R31A	stéллеllyey	stéллеlle	stéллеlloru
R31B	stéálellyey	stéálelle	stéálellor
R35B	stéálelyey	stéálele	stéálelor
R41A	stéálelyey	-----	-----
R41B	stéáleley	-----	-----
R53B	stéáleey	stéálele	stéálelor
R54	stéleyey	-----	-----
DR	stelei	stelele	stelelor

Again, haplology occurs to apocopate [stéleyey] to stelei in DR.

For a definitive explanation of the development of the Type II feminine, singular, nominative/accusative

enclitic article Sala refers us to Graur. As I will show below, this explanation also has bearing on the Type I article. I quote Graur [translation mine]:

A phonetic accident lead to the formation of a type of supplementary inflection: since Latin ll, in a post-tonic syllable, disappeared before a, feminine nouns [originally ending in -ella] came to end in stressed -a in the nominative/accusative singular, and in -le in the dative/genitive as well as in the plural (stea, stele). In Aromanian and Meglenoromanian, the corresponding nouns end in -eaua in the nominative/accusative singular (steaua). According to some researchers, this is the result of a normal evolution (ll > u); it is not known why the final -ua was lost in Daco-Romanian. According to others, when the article -a was postposed, a -u- was introduced between the final -a of the stem and the postposed article -a 'to hinder hiatus', or, more likely, to hinder the contraction of the two a's, which would have led to the confusion of the articulated form with the non-articulated form (stea-a > steaua). Later, in Aromanian, a back formation occurred which brought these nouns back in line with the majority of feminine nouns like masă/masa: steauă/steaua. This same back formation is found as a variant in some Daco-Romanian dialects: ziuă/ziua (instead of the standard zi/ziua), which is incontestably from the articulated form ziua. It is not clear why -ua was lost in some words and not in others such as piuă ['fulling mill' < L *pilla], rouă ['dew' < L ros, roris] and două ['two f' < L duae]. It is interesting to note that ella became ea in Aromanian, not eauă, while Latin mea, which never had -ll- became ameauă, clearly by analogy. If ella had become eauă, the feminine article would have had to be ieauă, therefore casa illa would have become caseauă, and stella illa, steauă-eauă" (Graur 1968:60-61).

It was statements like the ones above that lead me to try to put some order to the chaos by researching and writing this dissertation.

Let us attempt to derive both types of feminines organically, that is, by applying the language-general

historical phonological processes to the Latin etyma in order. For comparison we will derive the corresponding unarticulated forms and also the third person singular subject pronoun *ea* [ya] (< *ILLA*).

1. BR	élla	kása	kása élla	stélla	stélla élla	dí(e) dí élla	medúlla	medúlla élla
2. R29A	----	----	kása ella	-----	stélla ella	--- dí ella	-----	medúlla ella
3. R29B	----	----	kásalla	-----	stéllalla	--- dílla	-----	medúllalla
4. R30	yélla	----	-----	-----	-----	--- -----	-----	-----
5. R31A	yélla	----	-----	stélla	stéllalla	--- -----	-----	-----
6. R31B	yélla	----	-----	steálla	steállalla	--- -----	-----	-----
7. R35A	yéawa	----	kásawa	steáwa	steáwawa	--- díwa	medúwa	medúwawa
8. R36A	----	----	kásaa	-----	steáwaa	--- -----	-----	medúwaa
9. R36B	----	----	-----	-----	-----	--- díwaa	-----	-----
10. R36C	----	----	-----	-----	-----	--- -----	medúwa	medúwaa
11. R37A	yéawa	kásə	-----	steawa	-----	--- -----	medúwə	-----
12. R37B	-----	----	kása	-----	steáwa	--- díwa	-----	medúwa
13. R40	-----	----	----	-----	-----	--- dyíwa	-----	-----
14. R42A	-----	----	----	-----	-----	dzyí dzyíwa	-----	-----
15. R42B	-----	----	----	-----	-----	dzí dzyíwa	-----	-----
16. R46A	-----	----	----	-----	-----	-----	madúwa	madúwa
17. R52B	-----	----	----	-----	-----	zí zíwa	-----	-----
18. R57	yá	----	----	-----	-----	--- -----	-----	-----
19. DR	ea	casá	casa	stea	steaua	zi ziuu	máduvá	máduva
20.	'she'	'house'	'the house'	'star'	'the star'	'day' 'the day'	'marrow'	'the marrow'

If this derivation is correct, the historical basic form of the current DR feminine singular nominative/accusative article is -ua, of which -a is a variant. The -u- therefore would not be an infix which attached the unbound article [a] to a word ending in [ə] but rather the normal product of phonological rules applied to CL -LL- during the evolution of Romanian. That 'she' in DR is an apocopated ea ([ya]) and not eaua [yáwa] lends support to the idea that word-final [wa] had come to be identified as the feminine singular n/a definite article.¹⁵

CHAPTER 8

CONCLUSION

8.1 Synoptic Index of Sound Changes

It has been shown that sound changes can be described in terms of phonological rules which indicate very regular underlying developments. In conclusion I offer the following synoptic tabulation of the phonological rules formulated in the foregoing chapters accompanied by sample derivations (8.2) to further demonstrate their validity. The rules here are expressed in abbreviated form. The numbers in the center column refer to Otero's 'procesos fonológicos', and are included to enable the comparison of the sequencing of the early rules in Balkan- and Hispano-Romance.

R1	h > ø		hómo > ómo
R2A	ay > ε:	3B	káy lum > kÉ:lum
R2B	oy > e:		fóytus > fé:tus
R3A	m > ø / __#	6A	wétulum > wétulu
R3B	Vns > V:s	6B	ménsa > mé:sa
R4A	syncope	4	o:ríkula > o:ríkla
R4B	tl > kl		wétlu > wéklu
R5A	K > ø / __#	7	káput > kápu

R5B	∅ > e / l __#		mel > méle
R5C	metathesis of r#		ínter > íntre
R6	w > β	10	egzwolá:re > egzβolá:re
R7	y > dy / \$__		yurá:re > dyurá:re
R8	KsK > sK	8	egzβolá:re > ezβolá:re
R9	ŋkt > nt		úŋktu > úntu
R10A	rs > ss	13A	deórsu > deóssu
R10B	ps > ss	13B	skrípsi > skríssi
R10Ca	ks > ps kt > pt ŋn > mn		kóksa > kópsa ókto: > ópto: púŋnu > púmnu
R10Cb	ks > ss ŋn > nn		eksí:re > essí:re ko:ŋnó:skere > ko:nnó:skere
R11	k,g > p,b / __ wa		ákwa > ápwa
R12	prothesis	14	ská:la > iská:la
R13A	change of stress	15A	mulíere > mulié:re
R13B	e,o > i,u // V	15B	púteu > pútiu
R13C	V _i V _i > V _i	18B	mórtuu > mórtu
R13D	i,u > y,w // V	15C	pútyu > pútyu
R13E	é > ě	15Da	lépore > lěpore
R13F	i > e	15Db	kamísyā > kamésyā
R14	apheresis		eská:la > ská:la
R15	V contraction	18Aab	koórte > kó:rte
R16	loss of length	15F	ópto: > ópto
R17	w > ∅	T22R	kwéro > kéro
R18A	C _i > C _i C _i / __y	19	pútyu > púttyu

R18B	t,d > č,ǰ / __y	20Aa	púttyu > pútčyu
R18C	k,g > č,ǰ / __y	20Ab	brákkyu > brákčyu
R18D	y > ø after affric	20B	pútčyu > pútču
R18E	assim to palatal	20C	pútču > púčču
R19	depal after stress	32	púčču > púttu
R20	attrac of glide	23	róbbyu > róybbu
R21	b > β / V__	25	kabállu > kaβállu
R22A	β > v / #__	30	Beníre > veníre
R22B	β > b / C__		ferβente > ferbente
R22C	β > w / __R		stáβlu > stáwlu
R22D	β > g // u		úβer > úger
R22E	β > ø'		béβere > béere
R23	C _i C _j > C _i where C _i ≠ ll,nn,rr	28	addúko > adúko
R24	k,g > č,ǰ > __ e,i	21	kélu > čelu
R25A	s > y / __#		nós > nój
R25B	s > ø / __#		áwdis > áwdi
R25C	generalization of /-i/		skríes > skríi

COMMON ROMANIAN PERIOD (6th -11th centuries)

R26A	rr > trill		orresko > orresko
R26B	r > rr / #__		ríw > rríw
R27A	ɛ > e / __ NC		témpu > témpu
R27B	ɛ > ye	36	pétra > pyétra
R28	unstrs o > u		kándo > kándu
R29A	def art > -acc		lúpu éllu > lúpu ellu,

R29B apher/encl of def art	lúpu ellu > lúpullu
R30 e > ye / # ____	élle > yélle
R31A e,o > ɛ,ɔ / ____ Co a,e	negra > négra, kóda > kóda
R31B ɛ,ɔ > ɛa,ɔa	négra > neágra, kóda > koáda
R32 displacement of accent	méa > meá
R33 a > ə / ____ N o > u / ____ N	kánda > kóndu bónu > búnu
R34 l > r / V ____ V	soále > soáre
R35A lla > wa	steálla > steáwa
R35B ll > l	kaállu > kaálu
R36A wa > a in fem art	kásawa > kásaa
R36B a > aa in fem art (DR)	vúlpea > vúlpeaa
R36C uw > uv	medúwa > medúva
R37A a > ə	kása > kásə
R37B aa > a	kásaa > kása
R38 e > ɛ / ____ a	vúlpea > vúlpea
R39 i > e / [-acc]	citáte > cetáte
R40 i > yi	díku > dyíku
R41A s,l,n > š,λ,ñ / ____ y	yesyíre > yešyíre
R41B y > ø / s,λ,ñ ____	yešyíre > yešíre
R42A t,d > ts,dz / ____ y	tyéne > tsyéne
R42B y > ø	tsyéne > tséne
R43 kl,gl > kλ,gλ	óklu > ók u
R44 ə > e / ñ,λ ____	vyíñ > vyíñe
R45A sč > šč	sčíw > ščíw
R45B šč > št	ščíw > štíw

DACO-ROMANIAN PERIOD (11th century onward)

R46A e > ə / X __ Y	méru > mǎru, véntu > vǎntu
R46B əu > a / X __ Y	fǎátə > fáta, kǎməášə > kǎmášə
R47A ə > ɛ / __ NC	vǎntu > vǎntu
R47B e > i / __ N	byéne > byíne
R48 nn > n	pánnə > pánə
R49 ə > ɛ / __ r,s	vǎrtúte > vǎrtúte
R50 e,i > ə,ɛ / rr __	rréw > rrǎw, rríw > rrǐw
R51 rr > r	rrǎw > rǎw
R52A ʃ > ž / __ o,u	ʃukáre > žukáre
R52B dz > z	dzíce > zíce
R53A ɛa > a / š,ts __	šǎárpe > šárpe tsǎárə > tsárə
R53B ɛa > e / __Co e	meáse > mese
R54 λ > y	lépure > yépure
R55 ñ > y	spúñi > spúyi
R56 y > ø / __i	gǎyínə > gǎínə
R57 yǎa > ya	pyǎátrə > pyátrə
R58 iǎ > ie / __#	bukuríə > bukuríe
R59A i,u > [-voice] / __#	lúpi > lúpy, vekyu > vekyɸ
R59B w > ø / __#	vekyɸ > veky
R59C ii > i / __#	lúpii > lúpi
R60 ul > u / __#	kálul > kálu

8.2 Derivations

	ABSCONDERE	EXVOLĀRE	EXTRAPUNGERE
	abskóndere	egzβolá:re	ekstrapúngere
R8	askóndere	ezβola:re	estrapúngere
R14	-----	zβolá:re	strapúngere
R16	-----	zβoláre	-----
R22B	-----	zboláre	-----
R24	-----	-----	strapúnjere
R28	-----	zbuláre	-----
R33	askúndere	-----	-----
R34	-----	zburáre	-----
R37A	-----	-----	strəpúnjere
apocope	askúnde	zburá	strəpúnje
DR	a ascunde	a zbura	a străpunge
	'to hide'	'to fly'	'to pierce'

	ANNUM	PINNAM	ANGELUM
R3A	ánnu	pínna	ángelu
R4A	----	-----	-----
R13F	----	pénna	-----
R24	----	-----	ánĵelu
R31A	----	pénna	-----
R31B	----	peánna	-----
R33	----	-----	ónĵelu
R34	----	-----	ónĵeru
R37A	----	peánna	-----
R46A	----	peánna	-----
R46B	----	pánna	-----
R47A	----	-----	ínĵeru
R48	ánu	pánə	-----
R59	án	----	ínĵer
DR	an	pană	înger
	'year'	'feather'	'angel'

	BENE	BONUM	BONAM
	béne	bónum	bonam
R3A	----	bónu	bóna
R13E	béne	----	----
R27B	byéne	----	----
R33	-----	búnu	búna
R37A	-----	----	búnə
R47B	byíne	----	----
R56	bíne	----	----
R59	----	bún	----
DR	bine	bun	bună
	'well'	'good m sg'	'good f sg'

	CAELUM	QUAERŌ	GELUM
	káylum	kwáyro:	gelum
R2A	ké:lum	kwé:ro:	-----
R3A	ké:lu	-----	gélú
R13E	-----	-----	gélú
R16	kélu	kwéro	----
R17	----	kéro	----
R24	čélu	čéro	jéru
R27B	čyélu	čyéro	jyéru
R18D	čerú	čeró	jéru
R28	----	čerú	----
R59	čer	čer	jér
DR	cer	cer	ger
	'sky'	'I ask'	'frost'

	CAMISIAM	CALCANEUM	CAPITANEUM
	kamísiam	kalkáneum	kapitáneum
R3A	kamisia	kalkáneu	kapitáneu
R13B	-----	kalkániu	kapitániu
R13D	kamísyā	kalkányu	kapitányu
R13F	kamésya	-----	kapetányu
R31A	kamésya	-----	-----
R31B	kameášya	-----	-----
R33	-----	kalkényu	kapetányu
R37A	kəmeášyə	kəlkényu	kəpetányu
R41A	kəmeášyə	kəlkəñyu	kəpetəñyu
R41B	kəmeášə	kəlkəñu	kəpetəñu
R46A	kəməášə	-----	kəpə tə ñu
R46B	kəməášə	-----	-----
R47A	-----	kəlkəñu	kəpə tə ñu
R55	-----	kəlkéyu	kəpə tɛ yu
D59	-----	kəlkéy	kəpə tɛ y
DR	cămaşă	călcîi	căpătîi
	'shirt'	'heel'	'head of the bed'

	CANEM	PANEM	MANEM
	kánem	pánem	má:nem
R3A	kane	pane	má:ne
R16	----	----	máne
R33	kéne	péne	máne
R47A	kíne	píne	máne
diphthong	kýne	pýne	mýne
DR	cíne	píne	míne
	'dog'	'bread'	'morning'

	CATELLAM	MARGELLAM	VITELLAM	MEDULLAM
	katéllam	margéllam	vitéllam	medúllam
R3A	katélla	margélla	vitélla	medúlla
R13E	katélla	margélla	vitélla	-----
R24	-----	marjélla	-----	-----
R27B	katyélla	marjyella	vityélla	-----
R31A	katyeálla	marjyeálla	vityeálla	-----
R35A	katyeáwa	marjyeáwa	vityeáwa	medúwa
R36C	-----	-----	-----	medúva
reanalysis	katyeá	marjyeá	vityeá	-----
R37A	katyeá	mərjyea	-----	medúvə
R42A	katsyeá	-----	vitsyeá	-----
R42B	kətseá	mərjeá	vitseá	-----
R46A	-----	-----	-----	medúvə
DR	căṭea	mărgea	viṭea	măduvă
	'bitch'	'bead'	'heifer'	'marrow'

	CAUDAM	COXAM	DOMINAM
	kó:dam	kóksam	dóminam
R3A	kó:da	kóksa	dómina
R4A	-----	-----	dómna
R10Ca	-----	kopsa	-----
R16	kóda	-----	-----
R31A	kóda	kópsa	dómna
R31B	kṡáda	kṡápsa	dṡámna
R37A	kṡádə	kṡápsə	dṡámnə
DR	coadă	coapsă	doamnă
	'tail'	'thigh'	'lady'

	DECEM	DĪCERE	IACĒRE	IOCĀRE
	dékem	dí:kere	yaké:re	yoká:re
R3A	déke	-----	-----	-----
R7	----	-----	dyaké:re	dyoká:re
R13E	dĕke	-----	-----	-----
R16	----	díkere	dyakére	dyokáre
R18B	----	-----	ĵyakére	ĵyokáre
	----	-----	ĵakére	ĵokáre
R24	dĕće	díčere	ĵacére	-----
R27B	dyéće	-----	-----	-----
R28	----	-----	-----	ĵukáre
R31A	dyĕće	-----	ĵacére	-----
R31B	dyęáce	-----	ĵaceáre	-----
R37A	-----	-----	ĵećeáre	-----
apocope	-----	díče	ĵecea	ĵuká
R40	-----	dyíce	-----	-----
R42A	dzyęáce	dzyíce	-----	-----
R42B	dzeáce	dzíce	-----	-----
R52A	-----	-----	-----	žuká
R52B	zeáce	zíče	-----	-----
R53	zéče	-----	-----	-----
DR	zece	a zice	a zácea	a juca
	'ten'	'to say'	'to lie'	'to play'

Lausberg (1965:328) explians zácea (instead of the derived [ĵəčęá]) as a case of dissimilation.

	DENTEM	MENTEM	TEMPUM
	déntem	méntem	témpum
R3A	dénte	ménte	témpu
R13E	dĕnte	mĕnte	tĕmpu
R27A	dénte	ménte	témpu
R47B	dĭnte	mĭnte	tĭmpu
R59	-----	-----	timp
DR	dinte	mintе	timp
	'tooth'	'mind'	'time'

	DEORSUM	DEUM	DIGITUM
	deórsum	déum	dígitum
R3A	deórsu	déu	dígitu
R10A	deóssu	---	-----
R13B	dióssu	---	-----
R13D	dyóssu	déw	-----
R13E	-----	déw	-----
R13F	-----	---	dégetu
R18B	ǵyóssu	---	-----
R18D	ǵóssu	---	-----
R23	ǵósu	---	-----
R24	-----	---	déǵetu
R27B	-----	dyéw	-----
R31A	-----	-----	déǵetu
R31B	-----	-----	deǵetu
R42A	-----	dzyéw	-----
R42B	-----	dzéw	-----
R52A	ǵósu	-----	-----
R53B	-----	-----	déǵetu
R52B	-----	zéw	-----
R59	ǵós	---	déǵet
DR	jos	zeu	deget
	'down'	'god'	'finger'

	DIGNUM	SIGNUM	LIGNUM	PUGNUM
	díǵnum	síǵnum	líǵnum	púǵnum
R3A	díǵnu	síǵnu	líǵnu	púǵnu
R10Ca	dímnu	símnu	límnu	púmnu
R13F	démnu	sémnu	lémnu	-----
R59	démn	sémn	lémn	pumn
	'worthy'	'sign'	'wood'	'fist'

	FAMILIAM	GALLĪNAM	GLAKIAM
	famīliam	gallí:nam	glákiam
R3A	família	gallí:na	glákia
R13D	famīlya	-----	glákya
R13F	famélya	-----	-----
R16	-----	gallína	-----
R18A	-----	-----	glákkya
R18C	-----	-----	glákcya
R18D	-----	-----	glákča
R18E	-----	-----	gláčča
R19	-----	-----	gláttsa
R23	-----	-----	glátsa
R31A	famélya	-----	-----
R31B	fameálya	-----	-----
R35B	-----	galína	-----
R37A	fəmeálya	galínə	glátsə
R40	-----	gəlyínə	-----
R41A	fəmeálya	gəlyínə	-----
R41C	fəmeálə	gəlínə	-----
R43	-----	-----	glyátsə
R44	fəmeále	-----	glátsə
R53	fəméle	-----	-----
R54	fəméye	gəínə	gyátsə
DR	femeie	gəínă	gheaṭă
	'woman'	'hen'	'ice'

The first [e] of femeie is said to be due to assimilation to the other [e]'s.

	LAVĀRE	HABĒRE	BIBERE	VIDĒRE
	lawá:re	habé:re	bíbere	widé:re
R1	-----	abé:re	-----	-----
R6	laβá:re	-----	-----	βidé:re
R13F	-----	-----	bébere	βedé:re
R16	laβáre	abére	-----	βedére
R21	-----	aβére	béβere	-----
R22	-----	avére	-----	-----
R22A	-----	-----	-----	vedére
R22E	laáre	-----	beere	-----
R31A	-----	avére	beɾre	vedɾre
R31B	-----	aveáre	beḡáre	vedeḡáre
R37B	láre	-----	beáre	-----
apocope	lá	aveá	beá	vedeá
DR	a la	a avea	a bea	a vedea
	'to wash'	'to have'	'to drink'	'to see'

	LĪNUM	LĒGEM	LEPOREM	LAXĀRE
	lí:num	lé:gem	léporem	laksá:re
R3A	lí:nu	lé:ge	lépore	-----
R10Cb	-----	-----	-----	lassá:re
R13E	-----	-----	lépore	-----
R16	línu	lége	-----	lassáre
R23	-----	-----	-----	lasáre
R24	-----	leǵe	-----	-----
R27B	-----	-----	lyépore	-----
R28	-----	-----	lyépure	-----
R31A	-----	leǵe	-----	-----
R31B	-----	leǵǵe	-----	-----
R38	-----	-----	-----	lǝsáre
R40	lyínu	-----	-----	-----
R41A	lyínu	-----	lyépure	-----
R41B	línu	-----	lépure	-----
apocope	-----	-----	-----	lǝsá
R53B	-----	léǵe	-----	-----
R54	yínu	-----	yépure	-----
R56	ínu	-----	-----	-----
R59A	ín	-----	-----	-----
DR	in	lege	iepure	a lasa
	'flax'	'law'	'hare'	'to let'

	MENSAM	PENSUM	DENSUM
	ménsam	pénsum	dénsum
R3A	ménsa	pénsu	dénsu
R3B	mé:sa	pé:su	dé:su
R16	mésa	pésu	désu
R31A	mésa	-----	-----
R31B	mǵása	-----	-----
R37A	mǵásǝ	-----	-----
R46A	mǵásǝ	pásu	-----
R46B	másǝ	-----	-----
R59	-----	pǝs	des
DR	masǎ	pǎs	des
	'table'	'grievance'	'often'

	OCULUM	VETULUM	ASSULAM
	ókulum	wétulum	ássulam
R3A	ókulu	wétulu	ássula
R4A	óklu	wétlu	ássla
R4B	----	wéklu	áskla
R6	----	þéklu	-----
R22A	----	véklu	-----
R27	----	-----	-----
R37A	----	-----	ásklǝ
R43	ókλu	vékλu	ásklǝ
R44	----	-----	áskle
R54	ókýu	vékýu	áskye
R59	óký	véký	-----
DR	ochi	vechi	aschie
	'eye'	'old'	'splinter'

Vechi may derive from [wé:tulum] since it did not undergo diphthongization in R27.

	PETRAM	TERRAM	PECCĀTUM
	pétram	térram	pekká:tum
R3A	pétra	térra	pekká:tu
R13E	pétra	térra	-----
R16	-----	-----	pekkátu
R23	-----	-----	pekátu
R27B	pyétrá	tyétrá	-----
R31A	pyétrá	tyétrá	-----
R31B	pyeátra	tyeárra	-----
R37A	pyeátrǝ	tyeárrǝ	-----
R42A	-----	tsyeárrǝ	-----
R42B	-----	tseárrǝ	-----
R46A	-----	-----	pəkátu
R51	-----	tseárrǝ	-----
R53A	-----	tsárrǝ	-----
R57	pyátrǝ	-----	-----
R59	-----	-----	pəkát
DR	piatrǎ	ṭarǎ	păcat
	'stone'	'land'	'sin'

	QUĀLEM	QUANDŌ	SQUĀMAM
	kwá:lem	kwándo:	skwá:mam
R3A	kwá:le	-----	skwá:ma
	ká:le	kándo:	ská:ma
R12	-----	-----	eská:ma
R14	-----	-----	ská:ma
R16	kále	kándo	skáma
R28	----	kándu	-----
R33	----	kéndu	-----
R34	káre	-----	-----
R37A	----	-----	skáme
R47A	----	kéndu	-----
R59	----	kənd	-----
DR	care	cínd	scamá
	'which'	'when'	'lint'

	QUATTUOR	AQUAM	EQUAM	LINGUAM
	kwátt(w)or	ákwam	ékwam	língwam
R3A	-----	ákwa	ekwa	língwa
R5C	kwáttro	----	----	-----
R11	pwáttro	ápwa	épwa	límbwa
R13E	-----	----	épwa	-----
R13F	-----	----	----	lémbwa
R17	páttro	ápa	épa	lémba
R23	pátro	---	---	-----
R27B	-----	---	yépa	-----
R28	pátru	---	---	-----
R31A	-----	---	yépa	-----
R31B	-----	---	yəápa	-----
R37A	-----	ápə	yəápə	lémbə
R47B	-----	---	-----	límbə
R57	-----	---	yápə	-----
DR	patru	apă	iapă	limbă
	'four'	'water'	'mare'	'tongue'

	RESĪNAM	RĪVUM	HORRESCŌ
	resí:nam	rí:wum	horrésko:
R1	-----	-----	orrésko:
R3A	resí:na	rí:wu	-----
R6	-----	rí:βu	-----
R16	resína	ríβu	orrésko
R22E	-----	ríw	-----
R26A	-----	---	orrésko
R26B	rresína	rríw	-----
R28	-----	----	urrésku
R37A	rresín	----	-----
R40	rresyín	----	-----
R41A	rrešyín	----	-----
R41B	rrešín	----	-----
R50	rrəšín	rrɪw	urrəsku
R51	rəšín	rɪw	urósku
R59	-----	---	urésk
DR	răşină	rɪ	urăsc
	'resin'	'river'	'I hate'

	SCĪRE	CRĒSCŌ	CRĒSCIS	CRĒSCIT
	skí:re	kré:sko:	kré:skis	kré:skit
R5A	-----	-----	-----	kré:ski
R13F	-----	-----	kré:skes	kré:ske
R16	skíre	krésko	kréskes	kréske
R24	sčíre	-----	krésčes	krésce
R25C	-----	-----	krésči	-----
R28	-----	krésku	-----	-----
R31A	-----	-----	-----	krésce
R31B	-----	-----	-----	kreásce
apocope	sčí	-----	-----	-----
R45A	ščí	-----	krésčci	kreásce
R45B	ští	-----	kréšti	kreáste
R53B	-----	-----	-----	kréste
R59	-----	kresk	kréšty	-----
DR	a şti	cresc	creşti	creşte
	'to know'	'I grow'	'you grow'	'he grows'

	SELLAM	SEPTEM	SERPENS	SĒRAM	SALEM	SOLEM
	séllam	séptem	-----	sé:ram	sálem	sólem
R3A	sélla	sépte	sérpe	sé:ra	sále	sóle
R13A	sélla	sépte	sérpe	-----	-----	-----
R16	-----	-----	-----	séra	-----	-----
R27B	syélla	syépte	syérpe	-----	-----	-----
R31A	syélla	syépte	syérpe	síra	-----	sóle
R31B	syéalla	syéapte	syéárpe	seára	-----	sóále
R34	-----	-----	-----	-----	sáre	sóáre
R35A	syéawa	-----	-----	-----	-----	-----
reanalysis	syéa	-----	-----	-----	-----	-----
R37A	-----	-----	-----	seáro	-----	-----
R41A	šyeá	šyeapte	šyeárpe	-----	-----	-----
R41B	šeá	šeapte	šeárpe	-----	-----	-----
R53A	šá	šápte	šárpe	-----	-----	-----
DR	ša	šapte	šarpe	searā	sare	soare
	'saddle'	'seven'	'snake'	'even- ing'	'salt'	'sun'

	VĒNAM	VENTUM	VENIT	VĪNUM	VĪNEAM
	wé:nam	wéntum	wénit	wí:num	wí:neam
R3A	wé:na	wéntu	wénit	wí:nu	wí:nea
R5	-----	-----	wéni	-----	-----
R6	βé:na	βéntu	βéni	βí:nu	βí:nea
R13B	-----	-----	-----	-----	βí:nia
R13D	-----	-----	-----	-----	βí:nya
R13E	-----	βéntu	βéni	-----	-----
R13F	-----	-----	βéne	-----	-----
R16	βéna	-----	-----	βínu	βínya
R22A	véna	véntu	véne	vínu	vínya
R27A	-----	véntu	-----	-----	-----
R27B	-----	-----	vyéne	-----	-----
R37A	vénə	-----	-----	-----	vínyə
R40	-----	-----	-----	vyínu	vyínyə
R41A	-----	-----	-----	-----	vyíñyə
R41C	-----	-----	-----	-----	vyíñə
R44	-----	-----	-----	-----	vyíñe
R46A	vəná	véntu	-----	-----	-----
R47A	vɛná	vintu	-----	-----	-----
R47B	-----	-----	vyíne	-----	-----
R55	-----	-----	-----	-----	vyíye
R56	-----	-----	víne	vínu	víye
R59	-----	-----	-----	vín	-----
DR	vínă	vintu	vine	vin	vie [víye]
	'vein'	'wind'	'comes'	'wine'	'vineyard'

NOTES

1. Of course not all the Junggrammatiker were as rigorously faithful to the 'regularity hypothesis' as, say, Osthoff and Brugmann. Hermann Paul, for example, has enjoyed something of a re-appraisal certainly since Weinreich, Labov and Herzog's ground-breaking essay (Weinreich's last lecture, delivered in April, 1966, at Austin, Texas), for which see "Empirical Foundations for a Theory of Language Change," in Lehmann & Malkiel, Directions for Historical Linguistics, Austin: U. of Texas, 1968, pp. 97-195, and also Giorgio Graffi, "Luoghi comuni su Hermann Paul (e la scuola neogrammatica)."

2. Yakov Malkiel makes an effort to reinstate Richter in the history of the discipline in Richter, Kleinere Schriften..., Innsbruck, 1977.

3. According to Tekavčić (1972:197-8),
il fonema /m/ in posizione finale doveva essere articolato in modo indistinto, quasi come un soffio nasale fin dai tempi più remoti della storia del latino. Nelle iscrizioni pompeiane la /m/ finale cade in gran parte dei casi e anche nei testi latini tardi questo è uno dei fenomeni più frequenti, tanto che sarebbe banale citarne degli esempi. Infine, nessuna lingua romanza conserva la /m/ finale..., il che significa che il fenomeno è veramente antico e panromanzo..."

4. According to Väänänen (1974:98), certain phonetic conditions are particularly favorable to syncope: 1) the contiguity of an internal vowel with a resonant [r,l,m,n]. As evidenced by the Romance languages, the unstressed vowel fell regularly between r and m, r and d, l and d, l and p just as it did between two consonants with similar points of articulation, especially between s and t. 2) The close vowels i, e, and u are more susceptible to syncope than are o and especially a.

5. An alternative suggestion is that such neuters as fel, mel, sal, etc. were provided with a regular accusative in -em or at least with a final -e on the model of MARE, etc., via morphologic analogy.

6. According to Väänänen (1974:112), glides are so open, they are not effective syllable boundaries, and usually end up becoming more consonantal or disappearing.

7. Hammond (1976:66) writes that "apparently g was nasalized in Latin before the dental nasal sound written n, so that gn in such a word as agnus 'lamb' sounded like ngn in English 'hangnail'." This theory goes back at least as far as Sturtevant, Edgar H. 1940, with a reaction by Ralph Ward in *Language* 20 (1944):73-77, a summary comment by W. Sidney Allen in his Vox latina: the Pronunciation of Classical Latin, and a review by William Wyatt in Language 42 (1966):666f.

8. Vasiliu writes,

Schimbarea (39) ([gn] > [mn]) trebuie să fie de dată post-comună (caracterizează numai română și dialectele italiene de sud, cf. Rosetti, 1964:103): în celelalte limbi romanice, reflexul este o oclisivă nazală muiată, [n], sau o geminată [nn]: lat. lignum, signum > log. linu, sinu: it. legno, sp. leño, it. segno, sp. seña, port. senha, fr. signe. În dialectele italiene de sud: lat. agnum, ligna > aunu, leuna, deci rezultatul e o (semivocală) labială, [w] comparabilă cu labială (nazală) din română: lemn, semn (lat. lignum, signum). Credem că schimbarea (39) trebuie apropiată de (38a) ([ks] > [ps], [(s)s]) și (38b) ([kt] > [pt]) : în toate cele tre cazuri avem a face cu o mutație din seria velară în seria labială înaintea unei consoane dentale [t,s,n]" (1968:93-94).

Another possibility is that Romanian -mn- developed from Latin [gn], in which case [g] became a fricative before [n] as it did in Hispano-Romance (Cf. Otero's P12 legna > legna). Fricative [g] may have then vocalized to wau (Cf. P16 legna > lewna). The Balkan tendency to close wau to [v] or [ʃ] after a vowel (LAUDO >> AR alavdu 'I praise' cf. Sandfeld-Jensen 1930:102) could have lead to [vn] and hence to [mn] via anticipatory assimilation.

An additional change of this sort is that posited by Rohlf's for So. Ital. dialects (1966e:368-9) which involves two sorts of anaptyctic glide: either yod or wau. AGNUS could become awnu or avunu and thence amnu/amunu, which are the current forms in Calabria.

9. In Sardinian AQUA >> abba, EQUA >> ebba, QUATTOR

>> battor, QUINQUE >> kimbe, SEQUESTARE >> sebbestare, SILIQUA >> silibba, INGUINA >> imbena, LINGUA >> limba, SANGUEN >> samben, ANGUILLA >> ambidda, LANGUIDU >> lambiddu (Wagner [Paulis] 1984:224ff). Tuttle suggests that in proto-Sardinian akwa > agwa > abwa > abba (personal communication).

10. In the mid-thirties, Benvenuto Terracini addressed the coalescence of CL wau and b ("Di che cosa fanno la storia gli storici del linguaggio? Storia dei tipi BENIO e NERBA nel latino volgare," AGI 27:133-52, 28:1-31 & 134-50--reprnt. in Linguistica al bivio, Napoli: Guida, 1981, 175-231), in an anticipation of Martinet's fortis (post-consonantal and post-pausal) versus lenis (intervocalic) alternations. Such fortis renderings are known as "betacismo" and they account for his sub-title with benio = UENIO and nerba = NERUA (cf. App. Probi ALUEUS non albeus). Even Tuscan participated in this reanalysis, compare FABA > fava, but corbo 'crow' < CORUU, nerbo 'tendon', serbare < SERUARE, etc. (cf. Rohlfs 1966e:373f).

11. According to Rohlfs (1968:27 note),

Si pone qui il problema del perchè l'italiano (col romeno) concordi col resto della Romania nell'adozione dei vocaboli all'accusativo nel singolare, e al plurale continui invece le forme nominative. Problema che richiede una soluzione strutturale: data l'innovazione della caduta delle consonanti finali avvenuta nel tardo latino dell'Italia centrale (da cui deriva certo il romeno), si sarebbe avuta identità fra singolare e plurale in capra/capra(s), lupo/lupo(s). Durante il periodo in cui perdurò il sistema bicasuale (nominativo/accusativo) ci si dovè dunque accorgere della convenienza d'assumere le coppie capra/capre, e lupo/lupi come massimamente distintive. Una volta consolidatasi come caratteristica del femminile l'uscita -e, facile diveniva anche il trapasso da cane(s) a cani, analogico a lupi; mentre lo sviluppo di nocce doveva restare contrastato fra l'analogia a capre e quella a cani (sing. cane/noce, pl. cani/noci)."

And on the related subject of the second person singular marker in the present indicative Rohlfs states,

Già in antico però si addivenne a un'ulteriore formazione analogica, per cui l'i della coniugazione in i (dormi < dormi:s) è passato alle altre coniugazioni. Questa i è riscontrabile già nei più antichi testi letterari italiani (1968:248).

12. Sala writes,
 Dans les études consacrées à l'histoire du roumain, on insiste sur le fait que, dans les textes du XVIIe siècle, on rencontre, à côté de r, un r à plusieurs vibrations, noté soit par rr (arratǎ, rrapaus), soit par un signe spécial dérivé de l'r glagolitique (1976:76).

13. G. L. Trager (The Use of the Latin Demonstratives..., New York, 1932) reckons that the article appeared just after the seventh century and had existed in embryonic form in Latin...

14. The fall of -ll-, which is common to both Daco-Romanian and Aromanian, seems clearly to have occurred before the splitting of the dialects (Graur 1968:61).

15. In DR, unarticulated feminine nouns with stress on the final vowel form the plural by adding -le. This seems to be analogous to and stem from forms like stea/stele:

o zi	'a day'	o stea	'a star'
nişte zile	'some days'	nişte stele	'some stars'
ziua	'the day'	steaua	'the star'
zilele	'the days'	stelele	'the stars'
unei zile	'of a day'	unei stele	'of/to a star'
unor zile	'of some days'	unor stele	'of/to some stars'
zilei	'of/to the day'	stelei	'of/to the star'
zilelor	'of/to the days'	stelelor	'of/to the stars'

It is possible that words like zi aided in the establishment of -ua as the f sg n/a article.

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